

**My SubTitle**



**date and time**



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*rivt* is an open source software program for writing and formatting engineering documents as text, HTML or PDF files (referred to as *rivt docs*). A *rivt doc* is generated from a *rivt* file (.py) that includes *rivt* markup and imports the [rivtlib](#) Python package. Multiple *rivt docs* may be organized and linked as a *rivt* report. This site is an example of a *rivt report*.

An interface for finding public *rivt files* on *GitHub* is [here](#) and a repository of open source models and calculations, including *rivt*, is [here](#).

### A.1 Start

#### [1t] Run examples

An overview of *rivt* can be found [here](#). This section reviews how to get started in writing *rivt files* and publishing *rivt docs*. *rivt file* examples can be run in the cloud or on a local computer. In both cases editing and publishing is typically carried out in an IDE.

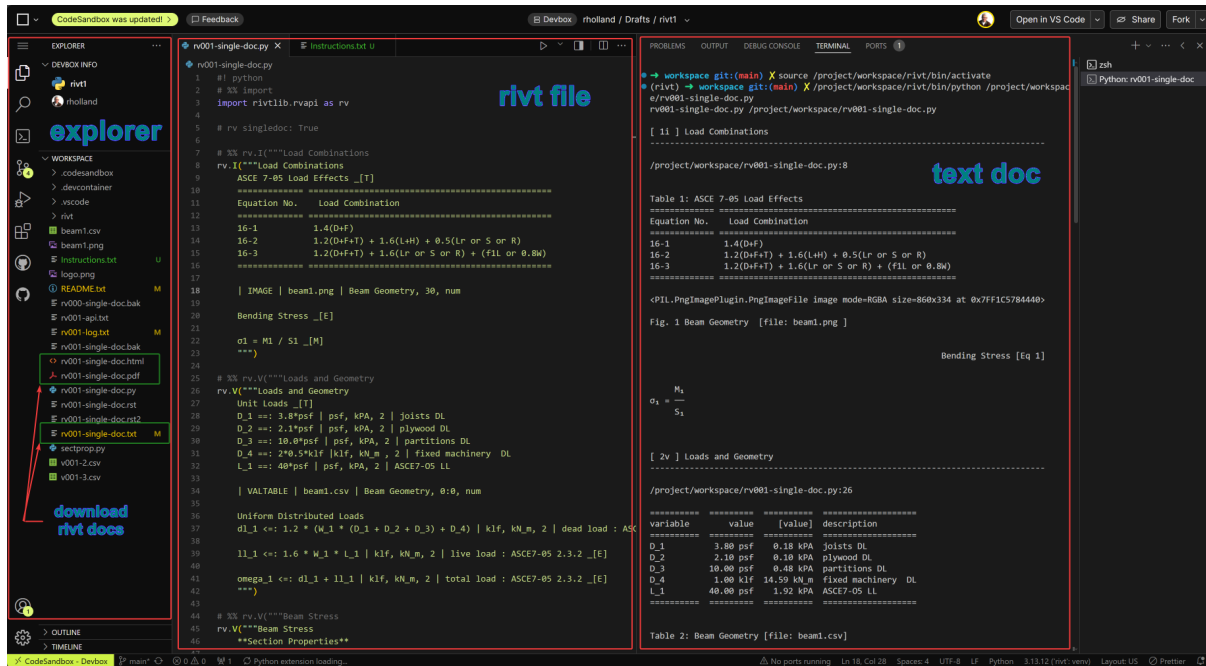
The three methods for installing and running *rivt* are:

1. *rivt-Codespace* in the cloud is described [here](#)
2. *rivt-portable* installation on Windows is described [here](#)
3. System installation with or without *uv* is described [here](#)

*rivt-portable* is recommended for Windows users unfamiliar with Python. *rivt-Codespace* is recommended when local installation is not practical or work is shared with teams. System installations are recommended for the most general and flexible implementation.

A typical **IDE** layout can be viewed at [Codesandbox](#).

## [2t] Codespace



This explorable browser example gives a feel for a rivt file (left panel) and text doc (right panel). Docs (output files) may be downloaded from the file explorer (far left panel) including the text, pdf and html reports.

## [2t] Codespace

VSCoDe can be run locally or in the cloud. For local installation see [5t] Framework.

The cloud version is a [Codespace](#). A rivt Codespace is a VSCoDe cloud environment with *rivt* extensions for editing and running *rivt* files. It can be forked (copied) into a personal Github account. Github setup is described [here](#).

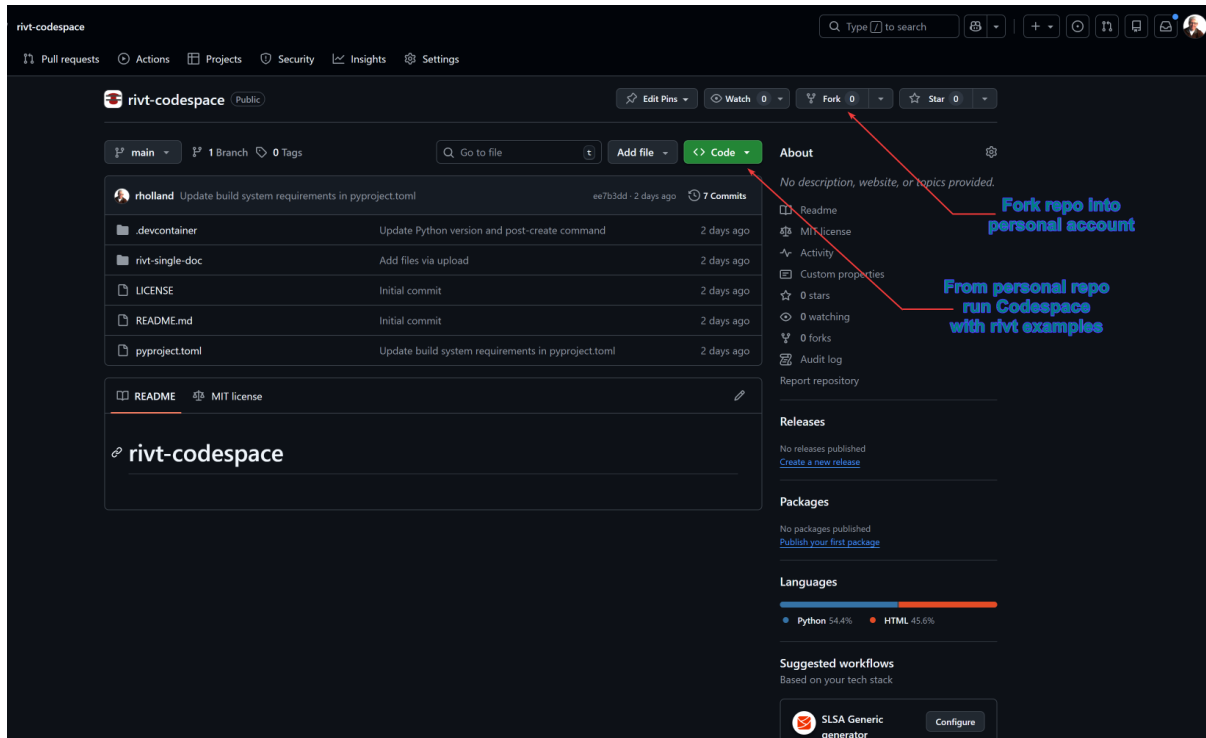
A rivt Codespace can be forked from [this repository](#) by following the steps below. Example rivt files are included.

### Fork a \*rivt Codespace\* in the cloud

1. Click the Fork button in the upper right corner.  
This creates a copy of the repository in your GitHub account.
2. Switch to your GitHub account and open the forked repository
3. Click the green Code button
4. Select the Codespaces tab
5. Click the rivt-codespace-examples option.
6. This creates a Codespace environment in your GitHub account and opens the repository in the online VSCoDe IDE. The environment includes the rivt packages and extensions needed to run and publish rivt docs and files.



## [3t] rvt-portable



### Local installation methods

1. Windows portable zip folder ([3t] rvt-portable) *rvt-portable* is recommended for Windows users unfamiliar with Python.
2. System Python ([4t] System and uv) with or without uv manager (rvt-uv)

### [3t] rvt-portable

A *rvt-portable* installation is recommended for Windows users unfamiliar with Python. *rvt-portable* is a zip file :

```
win64-rvt-portable-n.n.n[an].zip
```

where n is a number representing the major, minor and patch release number. An *an* appended to the version is an *alpha* release where users should expect that features are missing and *rvt markup* syntax may change in the future. Zip file contents must be unzipped into a directory with read-write access - typically the users home folder or a flash drive.

The zipped and unzipped file sizes include the following contents and are approximately 1GB and 2GB respectively:

```
release: rvt-portable-1.0.0a4.zip
```

1. Python 3.14 with rvt packages
2. VSCode 1.109 with rvt extensions

#### [4t] System and uv

3. `rivt-1.1.0a4`
4. Example `rivt` files

Releases may be download from the [GitHub repository](#).

The primary advantages of *rivt-portable* are

1. simplified installation
2. package integration
3. isolation from system files.

After unzipping, VSCode and examples are started with a single shortcut.

The primary disadvantages of *rivt-portable* are that *rivt* packages and extensions must be updated by updating the zip file as a whole, and it is more difficult to integrate with other Python programs.

#### [4t] System and uv

*rivt* may be installed at the system level.

1. Install Python using [Python installers](#) The minimum version required by *rivt* is Python 3.14.
2. Install *rivtlib* and dependencies using `pip`

```
pip install rivtlib
```

A list of the dependencies is here.

*rivt-uv*

Rather than installing *rivtlib* at the system level, it can be installed in an isolated environment using the [uv package manager](#). The primary advantage of *rivt-uv* is the simplicity of updating packages while keeping them isolated from the system Python.

*rivt-uv* installation is recommended for users with some familiarity with Python and programming.

##### Step 1. Install uv

Different methods for installing *uv* are described [here](#).

The recommended method for installing *uv* is from the command line:

Windows:

```
winget install --id=astral-sh.uv -e  
  
or  
  
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex
```

macOS and Linux:

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

## Step 2. Create the rvt environment

After installing *uv*, download and run the following *rvt* install script. It installs an isolated *rvt environment* and example files in a folder named *rvt-examples* in the users *Home* directory.

Windows: `rvtuv.cmd`

OSX and Linux: `rvtuv.sh`

## Step 3. Run the example program from Pyzo.

The *rvt environment* is activated by:

Windows:

```
./venv/Scripts/activate
```

macOS and Linux:

```
source venv/bin/activate
```

*rvt* includes the [Pyzo](#) IDE for editing and running examples. Typing the command *pyzo-example* from the environment root will load the example file *rv00-simple-doc.py* where it can be edited and run.

The example file can also be run from the command line with the command:

```
python -m rvt rv001-single-doc.py
```

The *uv* environment can be completely removed with the following commands.

```
uv deactivate
rmdir /s /q rvt-examples
uv cache clean
```

## A.2 Motivations

### [1t] Background

*rvt* is an open source software project that simplifies sharing and reuse of engineering documents. This has always been a challenge because the inclusion of text, images, tables, calculations, models and computer code required in reports has led to incompatible programs. Sharing and reuse has been restricted by the design and terms of use of existing software, including:

- documents written by different programs are incompatible
- frequent software updates are needed to maintain document access
- update costs are high
- newer document formats become inaccessible without upgrades
- software is limited to specific platforms
- document version control is limited
- report generation features are limited
- collaboration features are limited

The table below summarizes and compares limitations between different software programs. *rivt* is designed to address these limitations and function as a complement or replacement to existing software.

### Software Comparison

| Program     | Rep <sup>1</sup> | Ver <sup>2</sup> | Txt <sup>3</sup> | Comp <sup>4</sup> | CP <sup>5</sup> | Collab <sup>6</sup> |
|-------------|------------------|------------------|------------------|-------------------|-----------------|---------------------|
| Matlab      | no               | no               | no               | no                | no              | no                  |
| Mathcad     | no               | no               | no               | no                | no              | no                  |
| Mathematica | no               | no               | no               | no                | no              | no                  |
| Cloud SaaS  | limited          | no               | no               | no                | yes             | limited             |
| Excel       | limited          | no               | no               | yes               | no              | yes                 |
| Jupyter     | no               | no               | no               | yes               | yes             | yes                 |
| <b>rivt</b> | <b>yes</b>       | <b>yes</b>       | <b>yes</b>       | <b>yes</b>        | <b>yes</b>      | <b>yes</b>          |

- 1 Report generation
- 2 Native version control
- 3 Plain text, readable input files
- 4 Forward and backward compatibility
- 5 Cross-platform
- 6 Collaboration support

### [2t] Use Cases

The primary use case for *rivt* is producing engineering documents that lie somewhere between back of envelope notes and calculations, and formal journal publications. In other words, it produces formatted, organized documents that are easy to edit.

The second use case is when flexibility is needed to produce documents in a variety of formats including text, PDF or HTML.

*rivt files* can function as a front and back end for:

- 1. software control
- 2. visualization
- 3. instrumentation

*rivt docs* can be used for:

- 1. internal communication
- 2. research documentation
- 3. government permits
- 4. technical reports

## 5. funding applications

Because *rivt* is compatible with collaborative tools it may be used in:

1. teaching
2. for presentations
3. real time collaboration

## A.3 FAQ

### [1t] Contact

[support@rivt.info](mailto:support@rivt.info)

### [2t] Licenses

What is the open source license for *rivtlib*?

**rivtlib** is distributed under the [MIT open source license](#).

Which open source licenses applies to *rivt*?

The Python packages that are part of the *rivt project* are distributed primarily under the BSD, Appache and MIT open source licenses. Python itself is distributed under the Python Software Foundation License (PSFL).

Which open source licenses apply to *rivt files*?

The open source license for a public *rivt* file is specified in the `__[[Metadata]]` block tag. The default license is the MIT license.

Which open source licenses apply to the *rivt framework*?

**VSCode** - Visual Studio Code uses a proprietary Microsoft license for the compiled product, allowing free use for development, testing and commercial applications. The source code ("Code - OSS") is under the MIT license. You can use VS Code freely for any purpose, including commercial projects, but some bundled extensions or specialized components (connections to the Microsoft servers and marketplace) may have different terms. Most extentions in the Marketplace use the MIT license.

VSCodium is a fully open source alternative that removes Microsoft's proprietary components and built-in telemetry. However its extension marketplace is less complete.

**LaTeX** - LaTeX is released under the LaTeX Project Public License (LPPL) for LaTeX components and MIT and BSD variants for many core utilities and packages.

**QCAD** - The QCAD source code is distributed under the open source license GPLv3 with exceptions to allow proprietary extensions.

**Git** - The Git software is released under the GNU General Public License Version 2.0 (GPLv2).

### [3t] *rivt* design and development

What is the *rivt* release schedule?

1.0.x bug releases are issued as needed.

1.x.0 minor releases are planned monthly following the release of version 1.0.0

X.0.0. major releases will have backward compatibility. See [past releases](#) and [changelog](#) .

## B.1 Overview

### What are the programs and libraries used to build *rivt*?

The *rivt* project uses dozens of Python libraries (see [6t] Project requirements):

- numpy, sympy and related scientific computing libraries
- docutils and related text processing libraries
- Reportlab, rst2pdf and related PDF generation libraries
- Sphinx, Pydata and related HTML libraries

The *rivt* framework uses the following open source and free programs:

- [VSCode](#) for editing
- [LaTeX](#) for PDF generation
- [QCAD](#) for diagramming
- [Git](#) for document sharing and version control

## B.1 Overview

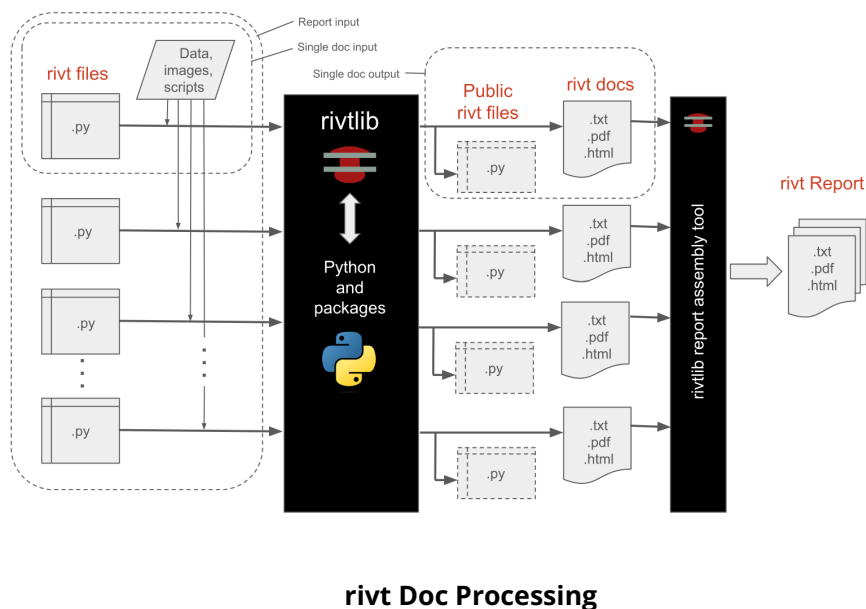
### [1t] Summary

*rivt* is an open source Python project that imports the [rivtlib Python package](#) and dependencies ([6t] Project requirements). The lightweight [Pyzo](#) IDE is also installed for editing and publishing *rivt* file examples. The [VSCode IDE](#) is a more full featured IDE that is part of the *rivt* framework and installed separately.

A *rivt* file publishes a formatted *rivt* doc as a text, PDF or HTML file. A *rivt* file is a Python file (.py) that imports the *rivtlib* Python package and includes *rivt* markup. A collection of *rivt* docs may be collated as a *rivt* report.

*rivt* file examples are provided [here](#). An interface for searching *public rivt* files on [GitHub](#) is [here](#). A *public rivt* file is all or parts of a *rivt* file the author chooses to share under an [Open Source license](#). *rivt* itself is distributed under the [MIT open source license](#). (see [2t] Licenses).

### [2t] rivt docs



## [2t] rvt API

Each rvt file outputs a corresponding doc in the format specified in *rv.D*. A doc number has the form:

```
rvAnn-filename.py
```

where *rv* is a required prefix, *A* is an alphanumeric character and *nn* are two digit non-negative integers.

A *rvt report* is organized using the *rvt doc numbers*. If the *rvt file* names are:

```
rvA01-filename.py  
rv105-filename.py  
rv212-filename.py
```

the *report numbers* would be:

- A.1 (division A, subdivision 1)
- 1.5 (division 1, subdivision 5)
- 2.12 (division 2, subdivision 12)

Note that leading zeroes are dropped. *Docs* are sorted alpha-numerically into divisions and subdivisions in the *report*.

*Docs* are assembled from sources and published using a specified folder structure. The *singledocB* variable overrides the default report structure and specifies that resource files and *docs* are read from and written to the *rvt file* folder. It is intended for simpler, standalone *docs* with more limited format control. The comment variable is specified immediately after the import statement:

```
import rivtlib.rvapi as rv  
  
# rv singledoc = True
```

The default setting is False.

## [2t] rvt API

The *rvt API* includes API functions, [3t] Markup and files. The API is designed to be:

- **lightweight**  
rvt markup is made up of about 3 dozen tags and commands, and wraps reStructuredText.
- **flexible**  
A *rvt file* produces a text, HTML or PDF *doc*. Multiple *docs* can be organized into a report.
- **extensible**  
*rivtlib* is written in Python with direct access to thousands of Python packages.

## [3t] Markup

An API function starts in the first column and takes a triple quoted rvt string (*rS*) argument. The first line of a *rvt string* is a header substring, followed by a content substring indented 4 spaces for improved readability and section folding.

The *header substring* specifies the section title and other processing parameters. The content substring includes rvt markup and other arbitrary text. For further details see [here](#).

```
rv._("""Section Label | parameters

    Content text that is
    indented four spaces.

    ...

""")
```

#### [4t] Files/Folders

A rvt file is a Python plain text file ( *.py* ) that includes *rvt markup* and imports the rvtlib package and API into the *rv* namespace:

```
import rvtlib.rvapi as rv
```

*rvt files* are stored in a root folder designated with the prefix *rvt-*. Each *rvt file* and corresponding *rvt doc* has a prefix that is used to organize the report. The prefix has the form

```
rvANN-filename.py
```

where A is an alpha-numeric character used for organizing report divisions, and NN is a two digit number used to organize sub-divisions. The file name is used as the *doc* title unless overridden in *rv.D*. Use underscores or hyphens rather than spaces to separate words in the file name. They are replaced with spaces when a *doc* title is extracted.

The top level folder structure is shown below. A more detailed description of the folder structure is here.

|                          |                            |
|--------------------------|----------------------------|
| [rvt-]Report-Label/      | Report Folder              |
| ├── [rv101-]filename1.py | rvt file                   |
| ├── [rv102-]filename2.py | rvt file                   |
| ├── [rv201-]filename3.py | rvt file                   |
| ├── [rv202-]filename4.py | rvt file                   |
| ├── ...                  |                            |
| ├── [.vscode]/           | optional VSCode settings   |
| ├── [_public]/           | public rvt files           |
| ├── [_publish]/          | published docs and reports |
| ├── [_stored]/           | stored files               |
| ├── [Rst_docs]/          | restructured text files    |
| ├── [Src]/               | source files               |
| └── README.txt           | searchable text report     |

#### [5t] Framework

The *rvt framework* includes separate programs that provide editing tools, additional formatting, version control, and diagramming.

| Purpose | Link                   |
|---------|------------------------|
| Editing | <a href="#">VSCode</a> |



| Purpose            | Link                    |
|--------------------|-------------------------|
| Editing Extensions | [7t] VSCode settings    |
| Added Formatting   | <a href="#">LaTeX</a>   |
| Version Control    | <a href="#">Git</a>     |
| 2D Diagramming     | <a href="#">QCAD</a>    |
| 3D Diagramming     | <a href="#">FreeCad</a> |

## C.1 Markup

### [1t] API functions

*rivt* has seven API functions. The name *rivt* is an acronym taken from the four functions that process content. The remaining three functions are used for document generation and debugging.

| API Function     | Name   | Purpose                               |
|------------------|--------|---------------------------------------|
| <b>rv.R</b> (rS) | Run    | Run shell commands                    |
| <b>rv.I</b> (rS) | Insert | Insert static sources                 |
| <b>rv.V</b> (rS) | Values | Calculate values                      |
| <b>rv.T</b> (rS) | Tools  | Python and markup scripts             |
| <b>rv.D</b> (rS) | Doc    | Publish docs                          |
| <b>rv.S</b> (rS) | Skip   | Skip section (comments and debugging) |
| <b>rv.X</b> (rS) | Exit   | Exit rivt (debugging)                 |

API functions define doc sections. If interactive IDEs like *VSCode* or *Spyder* are used to edit and run *rivt* files, functions can be processed individually as cells by using the standard notebook hash notation:

```
# %% optional label
rv.API_FUNCTION("""rivt string""")
```

### [2t] rivt string

Each API function takes a triple quoted rivt string argument composed of a line header substring line followed by a multiple-line content substring.

The *header substring* defines section processing parameters. The *content substring* includes rivt markup and is indented four spaces for improved readability and navigation (e.g. section folding).

#### Header substring

The header substring starts with a *section label* that is used as the section title, followed by a vertical bar and three comma separated *section parameters* that override default behavior.

```
rv._("""Section Label | doc;stored, private;public, section;merge
    Content substring indented 4 spaces
    ...
    """)
```

The parameters include the following, in any order:

***private/public***

Determines whether the API section text is copied to the the *Public* folder *rvt file* for sharing.

***doc/stored***

Determines whether the *rvt string* is formatted and printed in the doc, or just annotated in the doc and written to a file in the *Stored* folder for optional inclusion as an appendix.

***section/merge***

Determines whether the API starts a new *doc* section or is merged into the previous section.

Default settings in the *header substring* do not need to be specified. The default setting for each API is listed first (in bold) in the table below.

| API  | <b>private;public</b> | <b>doc;stored</b> | <b>section;merge</b> |
|------|-----------------------|-------------------|----------------------|
| rv.R | <b>private;public</b> | <b>stored;doc</b> | <b>merge;section</b> |
| rv.I | <b>private;public</b> | <b>doc;stored</b> | <b>section;merge</b> |
| rv.V | <b>private;public</b> | <b>doc;stored</b> | <b>section;merge</b> |
| rv.T | <b>private;public</b> | <b>stored;doc</b> | <b>merge;section</b> |
| rv.S | <b>private;public</b> | <b>stored</b>     | <b>merge</b>         |
| rv.D | <b>public</b>         | <b>stored</b>     | <b>merge</b>         |
| rv.X | •                     | •                 | •                    |

Examples of *header substring* settings are shown below.

- An example with explicit defaults (they do not have to be declared).

```
# This
rv.I("""A New Section | private, doc, section
    Content text
    ...
    """)
```

```
# is equivalent to:  
rv.I("""A New Section  
    Content text  
    ...  
    """)
```

- An example that merges a section into the previous section.

```
rv.I("""A Merged Section | merge  
    Content text  
    ...  
    """)
```

### Content substrings

The content substring is indented four spaces for legibility and code folding. It includes line tags, block tags and commands along with text.

```
rv._("""Section Label  
    Content text indented 4 spaces.  
    ...  
    """)
```

Content is converted line by line into formatted text and [RestructuredText](#), and then further processed into HTML or PDF. If a line does not contain a *command* or *tag* it is passed through as is, which allows for the *Insert* function (rv.I) including some *restructured text* directly i.e. surrounding words with \* or \*\* will format a word as italic or bold respectively.

In addition block tags in the *Tools function* (rv.T) directly supports processing HTML, LaTeX and reStructuredText scripts.

## [3t] Tags and Commands

### Line Tags

A line tag formats a line of text and is denoted with a single **\_[LETTER]**, placed at or near the end of the line, depending on the tag.

### Block Tags

A block tag formats a block of text and begins with **\_[TAGNAME]** and terminates with **\_[END]**.

## [4t] Folders

### Commands

*rivt* commands read and write external files. They typically start in the first column with a vertical bar ( | ) followed by the command name, file path, and parameters.

The exceptions are the definition (==:), assignment (<=:), function (:=:) and compare (<=>) commands that are used to define, assign and compare values.

```
| COMMAND | relative path | parameters
```

File paths are specified relative to the *rivt* root folder. The *rivt* report folder structure is described here..

If the path is omitted the default path for each command is applied. If the *singledoc* parameter is set, the *resource files* and *docs* are stored in the *rivt* root folder.

Tag and Command syntax for each API type is defined and described using the following format:

### [4t] Folders

For single docs the folder structure is the same as a *report* folder but simplified by omitting the *Src* and *\_stored* folders. The files normally stored there are stored in the root folder. The *report script* is not used for single docs.

The top level folder structure is shown below. A more detailed description of the folder structure is here.

```
[rivt-]Report-Label/      Report Folder
├── [rv101-]filename1.py   rivt file
├── [rv102-]filename2.py   rivt file
├── [rv201-]filename3.py   rivt file
├── [rv202-]filename4.py   rivt file
├── ...
├── [.vscode]/             optional VSCode settings
├── [_public]/             public rivt files
├── [_publish]/            published docs and reports
├── [_stored]/             stored files
├── [Rst_docs]/            restructured text files
├── [Src]/                 source files
└── README.txt            searchable text report
```

## C.2 Line Tags

### [1t] Line Tag Summary

#### Format a line of text

| API Scope | Line Tag | Description      |
|-----------|----------|------------------|
| rv.l      | text_[C] | [2t] Center text |

| API Scope | Line Tag                               | Description             |
|-----------|----------------------------------------|-------------------------|
| rv.l      | text <b>_R</b>                         | [3t] Right justify text |
| rv.l      | <b>text math _M</b>                    | [4t] Text math          |
| rv.l      | <b>LaTeX math _L</b>                   | [5t] LaTeX math         |
| rv.l      | text <b>_#</b> text                    | [6t] Endnote number     |
| rv.l      | text <b>_G glossary term</b>   text    | [7t] Glossary link      |
| rv.l      | text <b>_S label, section</b>   text   | [8t] Section link       |
| rv.l      | text <b>_D label, file name</b>   text | [9t] Doc link           |
| rv.l      | text <b>_U label, url</b>   text       | [10t] URL link          |
| rv.V, l   | text <b>_V var_name</b>   text         | [11t] Variable value    |
| rv.V, l   | <b>text _E</b>                         | [12t] Equation number   |
| rv.V, l   | <b>title _T</b>                        | [13t] Table number      |
| rv.V, l   | <b>caption _F</b>                      | [14t] Figure number     |
| rv.V, l   | <b>_P</b>                              | [15t] New Page          |
| rv.V, l   | <b>##</b> text                         | nonprinting comment     |

**[2t]** Center text

Center line of text within the page margins.

**\_C**

Syntax:

text **\_C**

Example:

This text wil be centered. **\_C**

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

**[3t]** Right justify text

Right justify line of text within the page margins.

`_R]`

Syntax:  
text `_R]`

Example:  
This text will be right justified `_R]`

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

#### [4t] Text math

Format math expression into text.

`_M]`

Syntax:  
text math expression `_M]`

Example:  
 $f(x,y) = \sin(x)**2 + y/5$  `_M]`

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

#### [5t] LaTeX math

Format LaTeX math expression.

`_L]`

Syntax:  
LaTeX math expression `_L]`

Example:  
 $\frac{1}{\sqrt{x}}$  `_L]`

|           |           |
|-----------|-----------|
| API Scope | Insert    |
| Doc Types | PDF, HTML |

### [6t] Endnote number

Assign endnote number to the text in order of processing. Endnotes are defined with the block tag `__[[ENDNOTE]]` and are listed at the end of the *doc*.

`_[#]`

Syntax:

text `_[#]` more text

Example:

This is a sentence with an endnote `_[#]` tag.

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

### [7t] Glossary link

Link a term to the glossary.

`_[G]`

Syntax:

text `_[G]` glossary term

Example:

A definition of `_[G]` namespace  
is provided in the glossary.

|           |           |
|-----------|-----------|
| API Scope | Insert    |
| Doc Types | PDF, HTML |

### [8t] Section link

Create a link to the section label defined in the API header with optional link text. If the text is omitted the section label is used for the link.

`_[S]`

**Syntax:**

text `_[S]` link text, section title

**Example:**

This creates a link to `_[S]` Section Title, actual section label which provides more detail.

|           |           |
|-----------|-----------|
| API Scope | Insert    |
| Doc Types | PDF, HTML |

#### [9t] Doc link

Link to a *doc* in a *report* with an optional link text. If the link text is omitted the *doc* title will be inserted as the link term.

`_[D]`

Syntax:

```
text _[D] text, rivt_file
```

Example:

```
This is a link to _[D] rv101-filename1  
for reference.
```

|           |           |
|-----------|-----------|
| API Scope | Insert    |
| Doc Types | PDF, HTML |

#### [10t] URL link

Link to a an external site with optional link text. If the link text is omitted the url will be inserted as the link term.

`_[U]`

Syntax:

```
text _[U] link label ,external url
```

Example:

```
text _[U] github, https://www.github.com
```

|           |           |
|-----------|-----------|
| API Scope | Insert    |
| Doc Types | PDF, HTML |

#### [11t] Variable value

Insert the value of `var_name _[V]` in the sentence.



**\_ [V]**

Syntax:

`text var_name _[V] more text`

Example:

`The value of my_var is my_var _[V].`

|           |                 |
|-----------|-----------------|
| API Scope | Insert, Values  |
| Doc Types | text, PDF, HTML |

**[12t]** Equation number

Assign equation number to a line of text.

**\_ [E]**

Syntax:

`math text or label _[E]``assignment command _[E]`

Example:

`3x * 4/(1+n) _[E]``x <= 3*IN + 4.1*FT | inch, cm, 2 | example 1 _[E]`

|           |                 |
|-----------|-----------------|
| API Scope | Insert, Values  |
| Doc Types | text, PDF, HTML |

**[13t]** Table number**\_ [T]**

Syntax:

`Table Title _[T]`

Example:

`A New Table _[T]`

|           |                 |
|-----------|-----------------|
| API Scope | Insert, Values  |
| Doc Types | text, PDF, HTML |

**[14t]** Figure number

\_[F]

Syntax:

Figure caption \_[F]

Example:

Stress Distribution \_[F]

|           |                 |
|-----------|-----------------|
| API Scope | Insert, Values  |
| Doc Types | text, PDF, HTML |

**[15t]** New Page

\_[P]

Syntax:

text \_[P]

Example:

Preceded by text or stand alone. \_[P]

\_[P]

|           |                 |
|-----------|-----------------|
| API Scope | Insert, Values  |
| Doc Types | text, PDF, HTML |

**C.3 Block Tags****[1t]** Block Tag Summary**Format blocks of text**

| API Scope | Block Tag                              | Description        |
|-----------|----------------------------------------|--------------------|
| rv.R      | _ <b>[[SHELL]]</b> process parameters  | [2t] Shell script  |
| rv.l      | _ <b>[[INDENT]]</b> spaces (4 default) | [3t] Indent text   |
| rv.l      | _ <b>[[ITALIC]]</b> spaces (4 default) | [4t] Indent italic |
| rv.l      | _ <b>[[ENDNOTES]]</b> optional label   | [5t] Endnotes      |
| rv.l      | _ <b>[[TEXT]]</b> optional language    | [6t] Text          |

| API Scope | Block Tag                                 | Description        |
|-----------|-------------------------------------------|--------------------|
| rv.l      | <code>_<b>[[TOPIC]]</b></code> topic      | [7t] Topic         |
| rv.l      | <code>_<b>[[BOX]]</b></code> label        | [8t] Box           |
| rv.V,T    | <code>_<b>[[PYTHON]]</b></code> namespace | [9t] Python        |
| rv.T      | <code>_<b>[[MARKUP]]</b></code> type      | [10t] Markup block |
| rv.D      | <code>_<b>[[METADATA]]</b></code> label   | [11t] Metadata     |
| rv.D      | <code>_<b>[[LAYOUT]]</b></code> label     | [12t] Layout block |
| all       | <code>_<b>[[END]]</b></code>              | [13t] End block    |

**[2t] Shell script**

Runs shell scripts that run external programs. The shell parameters include specifying the operating system, process control and terminal window control. The os parameter specifies the terminal type. The wait parameter specifies whether rvt file processing waits for the script to complete before continuing. The open parameter specifies whether to keep the shell window open after execution.

`_[[SHELL]]`

## Syntax:

```
_[[SHELL]] win;mac;linux,wait; nowait,open; close
  shell command
  ...
  _[[END]]
```

## Example:

```
_[[SHELL]] win,nowait,open
  dir
  path
  _[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

**[3t] Indent text**

Indents text block the specified number of spaces.

## \_[[INDENT]]

### Syntax:

```
_[[INDENT]] number of spaces  
text  
text  
...  
_[[END]]
```

### Example:

```
_[[INDENT]] 8  
This is a sentence that will be  
indented 8 spaces.  
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

## [4t] Indent italic

## \_[[ITALIC]]

Italicizes and indents the text block the specified number of spaces.

### Syntax:

```
_[[ITALIC]] number of spaces  
text  
...  
_[[END]]
```

### Example:

```
_[[ITALIC]] 4  
This is a sentence that will be  
italicized and indented 4 spaces.  
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

## [5t] Endnotes

## \_[[ENDNOTES]]

Assigns numbers and formats endnotes in order of processing. Each endnote is separated by a blank line and is numbered in order of occurrence.

Syntax:

```
_[[ENDNOTES]] label
Endnote 1.

Endnote 2.

...
_[[END]]
```

Example:

```
_[[ENDNOTES]] ACI citations
This endnote is assigned to the first endnote tag (_[#]) in order of
of processing.

This second endnote is separated by a blank line and assigned
to the second

This is a third endnote.
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

## [6t] Text

\_[[TEXT]]

Reads and formats text and code. The language parameter specifies formatting and syntax coloring. Languages include:

- *literal*
- *python*
- *text*
- *sh*
- *cmd*

Syntax:

```
_[[TEXT]] language
text and code

...
_[[END]]
```

Example:

```
_[[TEXT]] python
# some code
print(1)
a = 1 + 3
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

### **[7t]** Topic

`_[[TOPIC]]`

Formats a topic block.

Syntax:

```
_[[TOPIC]] topic title
text
...
_[[END]]
```

Example:

```
_[[TOPIC]] topic title
Text related to the topic.
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

### **[8t]** Box

`_[[BOX]]`

Draws a box around the block of text.

Syntax:

```
_[[BOX]] optional label
text
...
_[[END]]
```

Example:

```
_[[BOX]]
This is a sentence.
A second sentence.
A box is drawn around the sentences.
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Insert          |
| Doc Types | text, PDF, HTML |

**[9t]** Python`_[[PYTHON]]`

Executes Python script in the *rivt namespace* or a user specified namespace. File paths in the script are relative to the *rivt file* folder.

Syntax:

```
_[[Python]] topic title
code
...
_[[END]]
```

Example:

```
_[[PYTHON]] *rvspace*; user namespace
b = 1*inch + 2.2*ft
print(b)
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Value, Tool     |
| Doc Types | text, PDF, HTML |

**[10t]** Markup block

Inserts HTML into an HTML *doc*, LaTeX into a PDF *doc*, and reStructuredText into either PDF or HTML.

`_[[MARKUP]]`

Syntax:

```
_[[MARKUP]] *html;latex;rst*
markup
...
_[[END]]
```

Example:

```
_[[MARKUP]] rst
This is bold and
this is italic in
reStructuredtext
...
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Tool            |
| Doc Types | text, PDF, HTML |

**[11t]** Metadata

\_[[METADATA]]

*Metadata* is written to the *rvAnn-apilog.py* file and is converted internally to a Python dictionary *rvmetaD* with key name taken from the label e.g. *docnameS* or *authorsD*.

```
Syntax - defaults:
_[[METADATA]] optional label
docname = ""
authors = ""
version = 0.0.0
email = ""
repo = ""
license = https://opensource.org/license/mit
fork1 = author, version, email, repo
fork2 = author, version, email, repo
_[[END]]

Example:
_[[METADATA]]
docname = Bearing Pressures
authors = rholland, rward
version = 0.1.1
email = rholland@email.com
repo = https://github.com/rivt-info/rivt-simple-doc
license = https://opensource.org/license/mit
_[[END]]
```

|           |                 |
|-----------|-----------------|
| API Scope | Docs            |
| Doc Types | text, PDF, HTML |

**[12t]** Layout block

Overrides default layout settings.



`_[[LAYOUT]]`

Syntax - defaults:  
`_[[LAYOUT]]` optional label  
 variables  
 ...  
`_[[END]]`

Example:  
`_[[LAYOUT]]`  
`docname = ""`  
`footer = date, docname, page`  
`_[[END]]`

|           |                 |
|-----------|-----------------|
| API Scope | Doc             |
| Doc Types | text, PDF, HTML |

**[13t]** End block`_[[END]]`

Terminates a block tag.

Syntax:  
`_[[NEWPAGE]]` optional label

Example:  
`_[[END]]`

|           |                 |
|-----------|-----------------|
| API Scope | All             |
| Doc Types | text, PDF, HTML |

## C.4 Commands

**[1t]** Command Summary**format files and equations**

| API Scope | Command                            | Description     |
|-----------|------------------------------------|-----------------|
| rv.R      | <b>SHELL</b>   rel path   os, wait | [2t] Shell file |
| rv.l      | <b>TEXT</b>   rel path   language  | [3t] Text file  |

| API Scope | Command                                                      | Description          |
|-----------|--------------------------------------------------------------|----------------------|
| rv.V, I   | <b>TABLE</b>   rel path   title, width, rows, align, head    | [4t] Table file      |
| rv.V, I   | <b>IMAGE</b>   rel path   caption, scale, number             | [5t] Image file      |
| rv.V, I   | <b>IMAGE2</b>   rel pth1, rel pth2   cap1,cap2,sca1,sca2,num | [6t] Adjacent images |
| rv.V      | <b>VALTABLE</b>   rel path   title, rows, number             | [7t] Valtable file   |
| rv.V      | a ==: 1*IN   unit1, unit2, decimal   label                   | [8t] Define value    |
| rv.V      | c <=: expression   unit1, unit2, decimal   label             | [9t] Assign value    |
| rv.V      | c :=: expression   unit1, unit2, decimal   label             | [9t] Function value  |
| rv.V      | a < c   decimal   text1, text2, color1, color2               | [10t] Compare values |
| rv.T, V   | <b>PYTHON</b>   rel path   namespace                         | [11t] Python file    |
| rv.T      | <b>MARKUP</b>   rel path   type                              | [12t] Markup file    |
| rv.D      | <b>ATTACHPDF</b>   rel path   place, title                   | [13t] Attach PDF     |
| rv.D      | <b>PUBLISH</b>   doc title   type                            | [14t] Publish doc    |

### Default command paths

See here for the folder structure. If files are in the default path only the file name needs to be provided.

| Command   | Default Path         |
|-----------|----------------------|
| SHELL     | /Files/Scripts/      |
| TEXT      | /Files/Data/         |
| TABLE     | /Files/Data/         |
| IMAGE     | /Files/Image/        |
| IMAGE2    | /Files/Image/        |
| VALTABLE  | /Files/Data/ [1]     |
| PYTHON    | /Files/Scripts/      |
| MARKUP    | /Files/Scripts/      |
| ATTACHPDF | /Files/Attach/       |
| PUBLISH   | depends on file type |

[1] use /stored/data/filename to read values defined in the rivt file

**[2t]** Shell file

The SHELL command runs shell scripts including .cmd, .bat and .sh files. The *os* parameter specifies the operating system: *win*, *mac* or *linux*. The *wait*; *nowait* specifies whether rivt file processing waits for the script to complete before continuing.

If the *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/run/*. Otherwise the path is specified relative to the report root (rivt file folder). If the doc is a single doc the file is read from the rivt file folder.

**Syntax:**

```
| SHELL | relative file path | win;mac;linux,wait;nowait
```

**Example:**

```
| SHELL | file1.cmd | win, nowait
```

|            |                       |
|------------|-----------------------|
| API Scope  | rv.R                  |
| File Types | .cmd, .bat, .sh, .bsh |
| Doc Types  | text, PDF, HTML       |

**[3t]** Text file

The TEXT command reads and formats text and code files. The language parameter specifies formatting and syntax coloring. Language types include:

- *literal*
- *python*
- *text*
- *sh*
- *cmd*
- *reStructuredText*

The *literal* type inserts text into the *doc* without formatting.

If a *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/data/*. Otherwise the path needs to be specified relative to the report root (rivt file folder). If the doc is a single doc the file is read from the rivt file folder.

**Syntax:**

```
| TEXT | relative file path | language - see list above
```

**Example:**

```
| TEXT | file1.txt | literal
```

|            |                                 |
|------------|---------------------------------|
| API Scope  | rv.l                            |
| File Types | txt, .py, .cmd, .bat, .sh, .rst |
| Doc Types  | text, PDF, HTML                 |

**[4t]** Table file

The TABLE command reads csv, xls, and rst files and outputs formatted tables. The title may be omitted by inserting a hyphen "-". The width parameter specifies the maximum character width of a column. The align parameter specifies the cell justification - left, center, right. The number parameter specifies whether the table is numbered. For csv files, the head parameter specifies whether the first row is a column header.

If a *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/data/*. Otherwise the path needs to be specified relative to the report root (rivt file folder). If the doc is a single doc the file is read from the rivt file folder.

| TABLE |

Syntax:

| TABLE | rel path | title, max width, rows, l;c;r, head;nohead, num;nonum

Example:

| TABLE | file1.csv | Forces, 30, 0:0, c, nohead, num

|            |                 |
|------------|-----------------|
| API Scope  | rv.I, rv.V      |
| File Types | csv, xls, rst   |
| Doc Types  | text, PDF, HTML |

**[5t]** Image file

The IMAGE command reads a PNG or JPEG file and centers it in the *doc*. The scale parameter is a decimal fraction of the page width. The caption may be omitted by using a single hyphen. The *num;nonum* parameter specifies whether to assign a figure number. The image path is inserted in the text *doc* instead of the image.

If a *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/img/*. Otherwise the path needs to be specified relative to the report root (rivt file folder). If the doc is a single doc the file is read from the rivt file folder.

Syntax:

| IMAGE | relative path | caption, scale, num;nonum

Example:

| IMAGE | file1.png | Map, 0.5, num

|            |            |
|------------|------------|
| API Scope  | rv.V, rv.I |
| File Types | PNG, JPG   |
| Doc Types  | PDF, HTML  |

**[6t]** Adjacent images

The IMAGE2 command reads two PNG or JPEG file and places them side by side in the *doc*. The scale parameters are a decimal fraction of the page width. The captions may be omitted by using a single hyphen for either or both images. The *fig;nofig* parameters specify whether to assign a figure number to either or both images. The image path is inserted in the text *doc* instead of the image.

If a *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/img/*. Otherwise the path needs to be specified relative to the report root (rivt file folder). If the doc is a single doc the file is read from the rivt file folder.

**Syntax:**

```
| IMAGE2 | relative path | cap1, cap2, scale1, scale2, num;nonum, num;nonum
```

**Example:**

```
| IMAGE2 | file1.png, file2.png | Map, Photo, .5,.5, num, num
```

|            |               |
|------------|---------------|
| API Scope  | rv.I, rv.V    |
| File Types | PNG, JPG jenn |
| Doc Types  | PDF, HTML     |

**[7t]** Valtable file

The VALTABLE command imports and defines values from a csv or xls file. The file format is:

```
var = value, unit1, unit2, decimal, label
```

The path variable is either from *src* or *stored* depending on if the values were generated by rivtlib or provided by the author. The title parameter is the table title. A hyphen omits the title. The *rows* parameter specifies the rows to import. The *num; nonum* parameter specifies whether to assign a table number to the values table.

If a *doc* is part of a report and no path is specified, the file is assumed to be in the default folder */src/vals/*. Otherwise the path needs to be specified relative to the report root (rivt file folder). If the values are read from prior calculated values, they will be found in the */stored/vals* folder. If the doc is a single doc the file is read from the rivt file folder.

**Syntax:**

```
| VALTABLE | relative path | title, rows, *num;nonum*
```

**Example:**

If read from the default folder:

```
| VALTABLE | newvals.csv | Beam Properties, 1:10, num
```

If read from the stored folder:

```
| VALTABLE | /stored/vals/vA01-2.csv | Beam Properties, 1-10, num
```

|            |             |
|------------|-------------|
| API Scope  | rv.V        |
| File Types | .csv, .xls, |

|           |                 |
|-----------|-----------------|
| API Scope | rv.V            |
| Doc Types | text, PDF, HTML |

**[8t]** Define value

Defines a value and writes it to the file *vdocnum-s.csv* where *num* is the *docnumber* and *s* is the section number. The file is written to the folder *stored/vals* unless *singledocB* is set to *True* in the comment variable.

The stored values can read and defined in other rivt files using the VALUES command.

**Syntax:**

```
c =: 5*unit | unit1, unit2, decimals | label, *num,nonum*
```

**Example:**

```
D_1 =: 10*IN | IN, M, 3 | beam depth, num
```

|            |                 |
|------------|-----------------|
| API Scope  | rv.V            |
| File Types | .csv            |
| Doc Types  | text, PDF, HTML |

**[9t]** Assign value

Assigns a value to an equation and writes the values to a file *vdocnum-s.csv* where *num* is the *doc number* and *s* is the section number. The file is written to the folder *stored/vals* unless *rv single doc* is set to *True* in which case values are stored in the rivt file folder (root).

The label is a reference printed with the equation. The units specify the result expressed in two different units. The integer specifies the number of places after the decimals.

**Syntax:**

```
b <=: a * 10*FT | unit1, unit2, decimals | label, num;nonum
```

**Example:**

```
b_1 <=: E_1 * 12.1*IN^2 | KIP, KN, 2 | Std. 123, num
```

|            |                 |
|------------|-----------------|
| API Scope  | rv.V            |
| File Types | .csv            |
| Doc Types  | text, PDF, HTML |

**[9t]** Function value

Assigns a value to an equation and writes the values to a file *vdocnum-s.csv* where *num* is the *doc number* and *s* is the section number. The file is written to the folder *stored/vals* unless *rv single doc* is set to *True* in which case values are stored in the rivt file folder (root).

The label is a reference printed with the equation. The units specify the result expressed in two different units. The integer specifies the number of places after the decimals.

**Syntax:**

```
c :=: function name() | unit1, unit2, decimals | label, num;nonum
```

**Example:**

```
c_1 :=: func1(a,b) | KIP, KN, 2 | ACI 318-19 Table 22.5.5.1, num
```

|            |                 |
|------------|-----------------|
| API Scope  | rv.V            |
| File Types | .csv            |
| Doc Types  | text, PDF, HTML |

**[10t]** Compare values

Inserts a text with alignment and color on the following line if the expression evaluates to true. text1 and text2 are printed if true or false respectively. Colors are assigned accordingly as *red;blue;yellow;green;black;gray* defined in the *style files*. The text is right aligned.

Comparison operators:

```
==    Equal x == y
!=    Not equal x != y
>     Greater than x > y
<     Less than x < y
>=    Greater than or equal to x >= y
<=    Less than or equal to x <= y
```

<>

**Syntax:**

```
a < b | text1, text2, color1, color2
```

**Example:**

```
a < b | OK, NOT OK, blue, red
```

|           |                 |
|-----------|-----------------|
| API Scope | rv.V            |
| Doc Types | text, PDF, HTML |

**[11t]** Python file

Executes Python code in the *rvt namespace* or user specified namespace. File paths used in the script are relative to the root *rvt* folder.

Syntax:

| PYTHON | relative path | \*rvspace\*; user space

Example:

| PYTHON | script1.py | rv-space

|            |                 |
|------------|-----------------|
| API Scope  | rv.V, rv.T      |
| File Types | .csv            |
| Doc Types  | text, PDF, HTML |

**[12t]** Markup file

Inserts HTML into an HTML *doc*, LaTeX into a PDF *doc*, and reStructuredText into either PDF or HTML. LaTeX requires the installation of *texlive*.

| MARKUP |

Syntax:

| MARKUP | relative path | \*rst, html, latex\*

Example:

| MARKUP | script.rst | rst

|            |                 |
|------------|-----------------|
| API Scope  | rv.t            |
| File Types | html, rst, tex  |
| Doc Types  | text, PDF, HTML |

**[13t]** Attach PDF

Appends or prepends a PDF file to the *doc*. The title parameter generates an Appendix cover page with the specified title. A "-" omits the over page. For HTML *docs* the file is inserted as a download link.



| ATTACHPDF |

Syntax:

| ATTACHPDF | relative path | \*front;back\*, title

Example:

| ATTACHPDF | relative path | \*front;back\*, title

|            |                      |
|------------|----------------------|
| API Scope  | rv.D                 |
| File Types | tex, html, text, pdf |
| Doc Types  | text, PDF, HTML      |

**[14t]** Publish docPublishes a *doc*.

Syntax:

| PUBLISH | ini relative path | rst2pdf, texpdf, tex, hmtl

Example:

| PUBLISH | ini relative path | rst2pdf

|            |                      |
|------------|----------------------|
| API Scope  | rv.D                 |
| File Types | tex, html, text, pdf |
| Doc Types  | text, PDF, HTML      |

**C.5 Run - rv.R****[1i]** rv.R MarkupThe *Run* API function executes shell commands.**[2i]** Block Tags

| Block Tag                             | Description       |
|---------------------------------------|-------------------|
| _ <b>[[SHELL]]</b> process parameters | [2t] Shell script |
| _ <b>[[END]]</b>                      | [13t] End block   |

**[3i]** Commands

| Command                                 | Description     |
|-----------------------------------------|-----------------|
| <b>SHELL</b>   relative path   os, wait | [2t] Shell file |

## C.6 Insert - rv.l

### [1i] rv.l Markup

The *Insert* API function inserts and formats static sources into the *doc*, including images, tables, links and formatted text.

### [2i] Line Tags

| Line Tag                           | Description             |
|------------------------------------|-------------------------|
| text <b>_C</b>                     | [2t] Center text        |
| text <b>_R</b>                     | [3t] Right justify text |
| <b>text math _M</b>                | [4t] Text math          |
| <b>LaTeX math _L</b>               | [5t] LaTeX math         |
| text <b>_#</b> text                | [6t] Endnote number     |
| text <b>_G</b> text, section label | [8t] Section link       |
| text <b>_D</b> text, rvt_file      | [9t] Doc link           |
| text <b>_U</b> text, label         | [10t] URL link          |
| text <b>_V</b> var_name text       | [11t] Variable value    |
| assign var or <b>label _E</b>      | [12t] Equation number   |
| <b>title _T</b>                    | [13t] Table number      |
| <b>caption _F</b>                  | [14t] Figure number     |

### [3i] Block Tags

| Block Tag                             | Description        |
|---------------------------------------|--------------------|
| <b>_[[INDENT]]</b> spaces (4 default) | [3t] Indent text   |
| <b>_[[ITALIC]]</b> spaces (4 default) | [4t] Indent italic |
| <b>_[[ENDNOTES]]</b> optional label   | [5t] Endnotes      |
| <b>_[[TEXT]]</b> optional language    | [6t] Text          |

| Block Tag                     | Description     |
|-------------------------------|-----------------|
| <code>_[[TOPIC]]</code> topic | [7t] Topic      |
| <code>_[[END]]</code>         | [13t] End block |

**[4i] Commands**

| Command                                                           | Description          |
|-------------------------------------------------------------------|----------------------|
| <b>IMAGE</b>   relative path   scale, caption, figure             | [5t] Image file      |
| <b>IMAGE2</b>   rel path1, rel path2   s1, s2, c1, c2, fig1, fig2 | [6t] Adjacent images |
| <b>TEXT</b>   relative path   language                            | [3t] Text file       |
| <b>TABLE</b>   rel path   title, width, rows, align, head         | [4t] Table file      |

**C.7 Values - rv.V****[1i] rv.V Markup**

The *Values* API function defines values and evaluates equations and functions.

**[2i] Line Tags****Format a line of text**

| Line Tag                               | Description           |
|----------------------------------------|-----------------------|
| text <code>_[V] var_name</code>   text | [11t] Variable value  |
| <b>label</b> <code>_[E]</code>         | [12t] Equation number |
| <b>title</b> <code>_[T]</code>         | [13t] Table number    |
| <b>caption</b> <code>_[F]</code>       | [14t] Figure number   |

**[3i] Block Tags****Format or execute a block of text**

| Block Tag                          | Description     |
|------------------------------------|-----------------|
| <code>_[[PYTHON]]</code> namespace | [9t] Python     |
| <code>_[[END]]</code>              | [13t] End block |

**[2i]** Commands**Format files and calculations**

| Command                                                       | Description          |
|---------------------------------------------------------------|----------------------|
| <b>TABLE</b>   rel path   title,width,rows,align,head,num     | [4t] Table file      |
| <b>IMAGE</b>   relative path   scale, caption, figure         | [5t] Image file      |
| <b>IMAGE2</b>   rel path1, rel path2   c1, c2, s1, s2, f1, f2 | [6t] Adjacent images |
| <b>VALTABLE</b>   rel path   title, width, rows               | [7t] Valtable file   |
| <b>PYTHON</b>   relative path   namespace                     | [11t] Python file    |
| a ==: 1*IN   unit1, unit2, decimal, num   label               | [8t] Define value    |
| c <=: expression   unit1, unit2, decimal, num   label         | [9t] Assign value    |
| c :=: expression   unit1, unit2, decimal   label, number      | [9t] Function value  |
| a < c   decimal   text1,text2,align, num                      | [10t] Compare values |

**C.8 Tool - rv.T****[1i]** rv.T Markup

The *Tools* API function executes markup and scripts.

**[2i]** Block Tags

| Block Tag                    | Description        |
|------------------------------|--------------------|
| <b>_[[PYTHON]]</b> namespace | [9t] Python        |
| <b>_[[MARKUP]]</b> type      | [10t] Markup block |
| <b>_[[END]]</b>              | [13t] End block    |

**[3i]** Commands

| Command                                   | Description       |
|-------------------------------------------|-------------------|
| <b>PYTHON</b>   relative path   namespace | [11t] Python file |
| <b>MARKUP</b>   relative path   type      | [12t] Markup file |

## C.9 Doc - rv.D

### [1i] rv.D Markup

The *Doc API* publishes formatted *docs* from the rivt API strings.

### [2i] Block Tags

| Block Tag                 | Description        |
|---------------------------|--------------------|
| _ <b>[LAYOUT]</b> label   | [12t] Layout block |
| _ <b>[METADATA]</b> label | [11t] Metadata     |

### [3i] Commands

| Command                                         | Description       |
|-------------------------------------------------|-------------------|
| <b>PDFATTACH</b>   relative path   place, cover | [13t] Attach PDF  |
| <b>PUBLISH</b>   ini rel. path   type           | [14t] Publish doc |

## D.1 Docs

### [1t] rivt Files

Each rivt file outputs a corresponding formatted doc. A doc number has the form

```
rvAnn-filename.py
```

where *rv* is a required prefix, *A* is an alphanumeric character and *nn* are non-negative integers.

A *rivt report* is organized using the rivt *doc numbers*. If the *rivt file* names are:

```
rvA01-filename.py
rv105-filename.py
rv212-filename.py
```

the *report numbers* would be:

## [2t] Docs and Reports

- A.1 (division A, subdivision 1)
- 1.5 (division 1, subdivision 5)
- 2.12 (division 2, subdivision 12)

Note that leading zeroes are dropped. *Docs* are sorted alpha-numerically into divisions and subdivisions in the *report*.

### [2t] Docs and Reports

Each *rvt* file outputs a corresponding formatted doc written to the *publish* folder unless it is a single doc.

*Docs* may be text, HTML or PDF. PDF and HTML *doc* files are produced using the *rst2pdf* and *Sphinx* Python libraries. The *report folder* structure for *docs* is described in [3t] Report Folders. Reports are assembled from by the *rvt-report.py* script in the *Rst\_Docs* folder.

### [3t] Single docs

A single doc that will not be part of a report may use a simplified folder structure. A single *doc* is formatted by setting the comment variable, immediately following the *rvtlib* import statement.

```
from rvtlib import rvtlib.rvapi as rv
# rv singledoc=True
```

The folder structure for *single docs* is similar to the structure for *report* folders described in [3t] Report Folders but without the *Src* and *\_stored* folders. Files in those folders are stored or written in the root folder.

## D.2 Files

### [1t] File Types

#### **rvt file**

*rvt files* are written by the author. Each *rvt file* has a prefix that specifies the division label and subdivision number of the published *doc*. *rvt reports* are assembled using *doc numbers* that organize the report into divisions and subdivisions. The *d\*subdivision\** name is taken from the name that follows the *doc number*. It may be overridden in the *API Meta function*. *Division* names are specified in the report script.

The *doc number* prefix includes a single digit capital alphanumeric division label *D*, followed by a two digit subdivision number *ss*. For example:

```
[rvDss-]file-name.py
rvC02-Beam-Loads.py
```

where the default doc name is "Beam Loads" in subdivision 2 of division C. Hyphens are stripped from file name.

Note:

The following files are read and written by *rvtlib* when processing a *rvt file*. Files are labeled by origin and folder.

**origin:** author input

**folder:** *rivt-report-folder-name*

Refer to markup for *command* details.

### Run file

A *run file* is a shell or command script executed using the | RUN | command.

**origin:** author input

**folder:** *src/* - or a user created subfolder of */src*

**API function:** *rv.R*

### Insert file

An *insert file* is an image, table or text file executed using the | IMAGE | , | TABLE | , or | TEXT | command.

**origin:** author input

**folder:** *src/* - or a user created subfolder of */src*

**API function:** *rv.I*

### Value file

A *value file* is a *csv* file that defines variable values. It is imported using the | VALUE | command. Values defined by equations and a `_[[VALUES]]` block are written to a file with the name composed of the *doc* and *section* number. The file may be read by other *rivt files*.

Read

**origin:** author input

**folder:** *src/Values* - or a user created subfolder of */src*

**API function:** *rv.V*

Write

**origin:** program output

**folder:** *src/Values*

**API function:** *rv.V*

### Tool file

*Tool files* can be executed by | PYTHON | , | HTML | , | RST | or | LATEX | commands. They may be used for analysis and generation of text, tables, images and other formatted content. Executing a tool file may:

- define functions and classes for use in the *rivt file*
- generate images, tables and text for inclusion in the *doc*
- insert markup text into the *doc*

Read

**origin:** author input

**folder:** *src/Tools* or a user created subfolder of */src*

**API function:** *rv.T*

Write

**origin:** program output

**folder:** *src/Tools*

**API function:** *rv.T*

### Doc file

Each *rivt* file outputs a *doc* file with the same *doc name and number* and the selected *doc type* suffix. *Doc* files are written to the *publish folder* as text, PDF or HTML files using | PUBLISH | command.

PDF files may be prepended or appended to the *doc* using the | APPEND | command.

**origin:** program output

**folder:** *src/* - or a user created subfolder of */src*

**API function:** *rv.D*

### Report file

A *report file* is generated by the Python *report script* and is written to the designated *publish subfolder*. The *report file name* is taken from the *report folder name* unless specified in the *report script*.

**origin:** program output

**folder:** */publish/* subfolder

### Report script

The *report script* is a Python file that assembles the *report*. It specifies the *docs* that should be included, whether the *rivt* files should be re-executed and other format settings.

**origin:** author input

**folder:** */publish*

### Public rivt file

**origin:** program output

**folder:** */public*



A public rvt file is a copy of the *rvt file* that includes all of the sections marked as *public* in the *header*. It has the same name as the *rvt file* with an added hyphen between “rv” and the docnumber e.g. *rv-Dss-filename.py*

### Log file

*Log files* are written to the *log folder*. They include:

- a backup of the rvt file
- a file listing the execution steps when processing the *rvt file*
- a file listing each API function call with the header.

**origin:** program output

**folder:** */logs*

### [2t] Reports

A report is assembled from multiple *docs* organized by division and subdivision numbers. Each *doc* is a subdivision of a *report*. A typical workflow for writing *reports* is to start with a similar *report* and edit the *rvt files* and *report script* that produces the *report*.

The Python *report script* includes settings that specify assembly parameters and override defaults. These include:

- *Docs* to be included in the report
- Division names
- recompiling *docs*
- report cover page and title

provides the option to either regenerate all *docs* or to assemble the report from previously generated *docs*. Most aspects of the *report* appearance are determined at when generating *docs*.

An example script is shown below.

```
#!/ python

from rivtlib.rvreport import * # noqa: F403

""" generate a rvt report

Sample report generating and config file.

Run this Python file to generate a rvt report. The report generating file
must located in /publish/subfolder where subfolder is the doc type. The
report output file name is the report title. It is written to the same
folder as the report script. Duplicate report file names are incremented.

A flag determines whether the report is assembled from existing doc files
or whether docs are regenerated prior to assembly. If a rvt or doc file is
specified for inclusion and not found a warning is given but the report is
still assembled.

A rvt report is a compilation of rvt docs organized into divisions and
```

*subdivisions. rivt doc numbers define the organization. Default titles for subdivisions are generated by stripping the doc number from the file name and replacing hyphens with spaces. The default titles may be overridden in rv.M for each rivt file.*

*See src/styles/rivt.ini for additional settings*

```

"""
# =====
# report type
# =====
rptype = "pdf" # report type [html; pdf; pdftex; text]
rvrun = False # regenerate docs [True; False]

# =====
# report cover settings
# =====
# cover pages are located in /src
rptitle = "Solar Canopy Calculations"
rpstitle = "Larkspur, Ca"
rpauthor = "rhh"
rpdate = "<datetime>"
rptoc = True # add table of contents, "" to omit
rpcover = "/publish/cover1.pdf" # cover page, "" to omit

# =====
# include docs
# =====
# include these divisions in report
divlist = [[dv01, "Codes and Loads"],
            [dv02, "Frame"],
            [dv03, "Foundation"]]
# include these docs in report
# doc label is for reference only, may be empty string
doclist = [[rv101, "Codes"],
            [rv102, "Loads"],
            [rv201, "Frame"],
            [rv202, "Solar"],
            [rv301, "Foundation"],
            [rv302, "Walls"]]

genreport() # noqa: F405

```

## D.3 Folders

### [1t] File Names

A *rivt* report is assembled from a set of *docs*. Reports are organized using *doc numbers*. If the *rivt* file names are:

```

rvA01-filename.py
rv105-filename.py

```

The *report numbers* used in the published report would be:

```
A.1
1.5
2.12
```

where leading zeroes are dropped and *docs* are sorted alpha-numerically into divisions and subdivisions in the *report*.

Reports are organized under a single root report folder with the prefix *rivt-*. *rivt files* are stored in the root folder and *rivt markup* file paths are relative to the roo. Resource files are stored in five subfolders:

- \_public***  
Includes *rivt files* written by *rivtlib* intended for upload to a public repository. The prefix *rvAnn-* is changed to *rvAnn\_* to avoid confusion with private files.
- \_publish***  
Includes formatted *docs* and *reports* written by *rivtlib*.
- \_stored***  
Includes output files written by *rivtlib* including *logs*, *values*, *hidden*, and *metadata* and *reports*
- Rst\_docs***  
Includes formatted *docs* and *reports* written by *rivtlib*.
- Src***  
Includes author provided content, style and generating files for *docs* and *reports*.

An example *report* folder structure is shown below.

[3t] Report Folders

Folder Names

Top Level Folders

|                          |                            |
|--------------------------|----------------------------|
| [rivt-]Report-Label/     | Report Folder              |
| ├── [rv101-]filename1.py | rivt file                  |
| ├── [rv102-]filename2.py | rivt file                  |
| ├── [rv201-]filename3.py | rivt file                  |
| ├── [rv202-]filename4.py | rivt file                  |
| ├── ...                  |                            |
| ├── [.vscode]/           | optional VSCode settings   |
| ├── [_public]/           | public rivt files          |
| ├── [_publish]/          | published docs and reports |
| ├── [_stored]/           | stored files               |
| ├── [Rst_docs]/          | restructured text files    |
| ├── [Src]/               | source files               |
| └── README.txt           | searchable text report     |

Expanded Folders

|                          |                          |
|--------------------------|--------------------------|
| [rivt-]Report-Label/     | Report Folder Name       |
| ├── [.vscode]/           | optional VSCode settings |
| ├── [rv101-]filename1.py | rivt file                |

```

├── [rv102-]filename2.py      rivt file
├── [rv201-]filename3.py      rivt file
├── [rv202-]filename4.py      rivt file
├── [_public]/                public rivt files
│   ├── rv_101-filename1.py
│   ├── rv_102-filename1.py
│   ├── rv_201-filename3.py
│   └── rv_202-filename4.py
├── [_publish]/               published docs and reports
│   ├── [docs]/               html output
│   │   ├── _images/
│   │   ├── _sources/
│   │   ├── _static/
│   │   ├── process folders/
│   │   ├── site folders/
│   │   ├── rv101-filename1.html
│   │   ├── rv102-filename2.html
│   │   ├── rv201-filename3.html
│   │   └── rv202-filename4.html
│   ├── [pdfdocs]/            pdf doc output
│   │   ├── process folders/
│   │   ├── rv-101-filename1.pdf
│   │   ├── rv-102-filename1.pdf
│   │   ├── rv-201-filename3.pdf
│   │   └── rv-202-filename4.pdf
│   ├── [txtdocs]/            text output
│   │   ├── rv-101-filename1.txt
│   │   ├── rv-102-filename1.txt
│   │   ├── rv-201-filename3.txt
│   │   └── rv-202-filename4.txt
├── [_stored]/                stored files from rivt
│   ├── [Logs]/               log files
│   │   ├── rv101-log.txt
│   │   └── rv102-log.txt
│   ├── [Sect]/               stored sections
│   │   ├── rv202-5d.txt
│   │   ├── rv103-4t.txt
│   │   └── rv301-2r.txt
│   ├── [Temp]/               temp files
│   │   └── rv101-label3.tex
│   ├── [Vals]/               stored value files
│   │   ├── v101-2.csv
│   │   └── v102-3.csv
├── [Rst_docs]/               restructured text files
│   ├── _downloads/
│   ├── _static/
│   ├── _locale/
│   ├── _templates/
│   ├── rv101-filename1.rst
│   ├── rv102-filename2.rst
│   ├── rv201-filename3.rst
│   └── rv202-filename4.rst

```

|                              |                          |
|------------------------------|--------------------------|
| ├── gen-html.cmd             | html generating script   |
| ├── gen-pdf.cmd              | pdf generating script    |
| ├── gen-pdftex.cmd           | LaTeX generating script  |
| ├── rivt-report.py           | report generating script |
| ├── new-units.py             | define new units         |
| ├── coverpage.rst            | cover page template      |
| ├── logoname.png             | cover page logo          |
| ├── conf.py                  | style paths and settings |
| ├── [Src]                    | source files from author |
| │   ├── data/                | author created subfolder |
| │   │   ├── data1.csv        |                          |
| │   │   ├── opensees1.txt    |                          |
| │   │   └── conc-vals.csv    |                          |
| │   ├── image/               | author created subfolder |
| │   │   ├── fig1.png         |                          |
| │   │   └── fig2.jpg         |                          |
| │   ├── [Run]/               | OS commands              |
| │   │   ├── run1_win.cmd     |                          |
| │   │   └── run1_linux.sh    |                          |
| │   ├── [Tools]/             | scripts and functions    |
| │   │   ├── plot.py          |                          |
| │   │   └── loads.py         |                          |
| │   ├── [Vals]/              | value files              |
| │   │   ├── steel-vals.csv   |                          |
| │   │   └── plastic-vals.csv |                          |
| └── README.txt               | searchable text report   |

## D.4 Configuration Files

### [1] Config Files

### D.4 Example

#### [1] rivt file

This is a small rivt file that publishes a single *doc*. The report folders are not used. It has 4 API functions - two of them are published. The output is written to the terminal.

```
#!/python
import rivtlib.rvapi as rv

# rv singledoc: True

# %% loads
rv.I("""Load Combinations

ASCE 7-05 Load Effects _[T]
=====
Equation No.    Load Combination
```

```

=====
16-1          1.4(D+F)
16-2          1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3          1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)
=====

| IMAGE | beam.png | Beam Geometry, .65, num

Bending Stress Formula _[E]
o1 = M1 / S1 _[M]

"""
# %% values
rv.V("""Loads and Geometry

Beam Loads _[T]
D_1 =: 3.8*psf | psf, kPA, 2 | joists DL
D_2 =: 2.1*psf | psf, kPA, 2 | plywood DL
D_3 =: 10.0*psf | psf, kPA, 2 | partitions DL
D_4 =: 2*0.5*klf | klf, kN_m , 2 | fixed machinery DL
L_1 =: 40*psf | psf, kPA, 2 | ASCE7-05 LL

| VALTABLE | beam1-v.csv | Beam Geometry, 0:0, num

Uniform Distributed Loads
dl_1 <=: 1.2 * (W_1 * (D_1 + D_2 + D_3) + D_4) | klf, kN_m, 2 | dead load : ASCE7-05 2.3.2 _[E]

ll_1 <=: 1.6 * W_1 * L_1 | klf, kN_m, 2 | live load : ASCE7-05 2.3.2 _[E]

omega_1 <=: dl_1 + ll_1 | klf, kN_m, 2 | total load : ASCE7-05 2.3.2 _[E]

""")
# %% section properties
rv.V("""Beam Section Properties

| PYTHON | sectprop.py | nodocstring

section_1 <=: rectsect(10*inch, 18*inch) | in3, cm3, 2 | function: rect sect modulus _[E]

inertia_1 <=: rectinertia(10*inch, 18*inch) | in4, cm4, 1 | function: rect moment inertia _[E]

""")
# %% force
rv.V("""Force and Stress

m_1 <=: omega_1 * S_1**2 / 8 | ftkips, mkN, 2 | mid-span UDL moment _[E]

fb_1 <=: m_1 / section_1 | psi, MPA, 1 | bending stress _[E]

fb_1 < 10*ksi | ok, not ok, blue, red, right _[E]

""")

```

```

# %% tool
rv.T("""Metadata

_ [[PYTHON]]
rv_metaD = {
"authors": "rholland",
"version": "0.7.2",

```

```
"email": "rod.h.holland@gmail.com",
"repo": "https://github.com/rivt-info/rivt-single-doc",
"license": "https://opensource.org/licenses/mit/",
"fork1": ["author", "version", "email", "repo"],
"fork2": [],
}
_][[END]]

"""
# %% doc
rv.D("""Publish Doc

_][[LAYOUT]]
    logopathP: logo.png
    pdfheaderL: [docnameS, pageS, totalpageS]
    footerL: [logo0, date0, time0, filenameS, versionS]
_][[END]]

| PUBLISH | rivt | rst2pdf

""")
```

### D.5 Example

rivt files here

## E.1 Collaboration

### [1t] Overview

1. Public rivt files **may be shared by uploading** to repositories such as *GitHub*.
2. **Public *rivt* files can be found and downloaded using the** search interface. For public files,

- cloned versions may be forked to create different versions.
- pull requests may provide improvements.
- issues and bugs may be reported.

*Public rivt files* are identified with the naming convention `rv-A01-filename.py`, in contrast with `rvA01-filename` for private files. After downloading a public rivt file, use the *Batch Rename* extension to delete the hyphen after *rv* and convert it to a private rivt file.

3. Files may be collaboratively edited in real time using VSCode with the extension [Visual Studio Live Share](#) extension.
4. Contribute to rivtlib at [rivtlib-dev](#) and documentation at [rivt-info](#).
  - report issues and bugs
  - issue pull requests to propose changes

- contribute to documentation

5. **Contribute extensions and scripts that improve interaction *rikt* with other components of the *rikt framework*.**

**[2t] Git and GitHub**

[Git](#) is part of the *rikt framework*. It is a free, open-source, distributed version control system designed to manage and track changes in files. It allows multiple people to work on the same project simultaneously without overwriting each other's work.

[GitHub](#) is also part of the *rikt framework*. It is a web-based platform for hosting, managing, and collaborating on code built around Git. It allows teams to work together efficiently on software projects and provides features like pull requests, issue tracking, and project management.

Each *rikt project* is typically stored in its own repository. Every user of GitHub has a personal account with essentially unlimited repositories. Free accounts provide for:

- Unlimited Repositories: You can create as many public and private repositories as you need.
- Unlimited Collaborators: There is no limit to the number of people you can work with on your repositories.
- Community Support: You have access to the GitHub Community Discussions for help.
- Core Services with Usage Limits: The free plan includes a certain amount of monthly usage for services:
  - **GitHub Actions: 2,000 minutes per month for private repositories**  
(unlimited for public repositories).
  - GitHub Codespaces: 120 core hours and 15 GB of storage per month.

*GitHub Codespaces* is a VSCode cloud-based development environment connected to GitHub files. It can be set up with *rikt extensions* that provide a complete *rikt* editing and collaboration environment in the cloud.

**[3t] VSCode**

[VSCode](#) is an open source IDE with a large set of extensions, including collaboration support. Collaboration is facilitated by the [Visual Studio Live Share](#). You can start a *Live Share* session either within *CodeSpaces* in your browser, or within the *VS Code* desktop application. *CodeSpaces* is a cloud based GitHub implementation of VSCode that shares many of the same features.

The *Live Share* extension enables:

- Real-time Co-editing: Multiple participants can work on the same file simultaneously, seeing each other's cursors, selections, and edits instantly, much like in Google Docs..
- Shared Terminals and Servers: The host can share their terminal and localhost servers with guests, eliminating the need for guests to set up their own environments or dependencies.
- Navigation and Following: Participants can independently navigate the project or choose to "follow" another collaborator, keeping their editor synchronized with the leader's actions.
- File Access Control: Hosts can use a `.vscode.json` file to control which files and folders are visible or editable by guests, ensuring security and focus



## B.4 Git

### [1t] Git

*GitHub* is a popular and widely used cloud and version control system based on the Git program that may be downloaded [here](#).

## E.3 GitHub Search

### [1t] GitHub Global Search

This page is a search interface for **rivt** files on GitHub. The search covers the full text of the **rivt** README.txt file in the *repo* directory. The README file contains the *rivt report*. For advanced searching use the [GitHub interface](#).

[ S ] Search GitHub <Ctrl+Enter>

[ C ] Clear input <Ctrl+R>

**Example:** solar+steel+frame is passed to GitHub as rivt+solar+steel+frame

### [2t] GitHub Organization Search

Restrict search to the following comma separated GitHub organizations:

## E.2 Public rivt files

### [1t] Overview

A feature of *rivt* is fine-grained control over public sharing of all or parts of a rivt file. *Public rivt files* are generated by *rivtlib* on a section by section basis defined by the API header substring. Each API function defines a section and has a public/private setting in the header. Sections are private by default (see *:doc: <rvF01-quickref>*).

```
rv._("""Section Label | doc;stored, private;public, section;merge
    Content substring indented 4 spaces
    ...
    """)
```

Sections are written to a rivt file in the */public* folder with a file name composed as:

```
rivt file name
    rvAnn-filename.py
modified with an additional hyphen
```

```
rv-Ann-filename.py
```

The public folder may be uploaded to a GitHub repo for public sharing. When downloaded the extra hyphens may be removed with a batch remove command (e.g. in VSCode).

### [2t] Public Folder

For each rivt file that is published a corresponding *public rivt file* is written to the *public* folder. The default contents of the file are the API rivt string headers. If the header parameter *public* is included in the header, the rivt string content is also written to the *public rivt file*.

```
[rivt]-Report-Label/
├── [rv101-]filename1.py    | rivt file
├── [rv102-]filename2.py    | rivt file
├── [rv201-]filename3.py    | rivt file
├── [rv202-]filename4.py    | rivt file
...
└── [public]/              || public rivt files
    ├── rv-101-filename1.py
    ├── rv-102-filename1.py
    ├── rv-201-filename3.py
    └── rv-202-filename4.py
```

The *public folder* can be uploaded to a public repository and updated with Git as the project develops.

### [3t] Example

## F.1 Quick Ref

### [1t] API Summary

#### API functions

| Function         | Name   | Purpose                               |
|------------------|--------|---------------------------------------|
| <b>rv.R</b> (rS) | Run    | Run shell commands                    |
| <b>rv.I</b> (rS) | Insert | Insert static sources                 |
| <b>rv.V</b> (rS) | Values | Calculate values                      |
| <b>rv.T</b> (rS) | Tools  | Python and markup scripts             |
| <b>rv.D</b> (rS) | Doc    | Publish docs                          |
| <b>rv.S</b> (rS) | Skip   | Skip section (comments and debugging) |

## [2t] Line Tags

| Function  | Name | Purpose               |
|-----------|------|-----------------------|
| rv.X (rS) | Exit | Exit rivt (debugging) |

where **rS** is a *rivt string*. The first line of a *rivt string* (rS) is the header substring.

```
rv.I("""A New Section | private, doc, section
    Content text
    ...
    """)
```

The default setting for each API is listed **in bold** in the table below. Default settings in the *header substring* do not need to be specified.

| API  | private;public         | doc;stored         | section;merge         |
|------|------------------------|--------------------|-----------------------|
| rv.R | <b>private</b> ;public | <b>stored</b> ;doc | <b>merge</b> ;section |
| rv.I | <b>private</b> ;public | <b>doc</b> ;stored | <b>section</b> ;merge |
| rv.V | <b>private</b> ;public | <b>doc</b> ;stored | <b>section</b> ;merge |
| rv.T | <b>private</b> ;public | <b>stored</b> ;doc | <b>merge</b> ;section |
| rv.S | <b>private</b> ;public | <b>stored</b>      | <b>merge</b>          |
| rv.D | <b>public</b>          | <b>stored</b>      | <b>merge</b>          |
| rv.X | •                      | •                  | •                     |

## [2t] Line Tags

### Format a line of text

| API Scope | Line Tag                                     | Description             |
|-----------|----------------------------------------------|-------------------------|
| rv.l      | text_ <b>[C]</b>                             | [2t] Center text        |
| rv.l      | text_ <b>[R]</b>                             | [3t] Right justify text |
| rv.l      | <b>text math</b> _ <b>[M]</b>                | [4t] Text math          |
| rv.l      | <b>LaTeX math</b> _ <b>[L]</b>               | [5t] LaTeX math         |
| rv.l      | text_ <b>[#]</b> text                        | [6t] Endnote number     |
| rv.l      | text_ <b>[G]</b> <b>glossary term</b>   text | [7t] Glossary link      |

| API Scope | Line Tag                                 | Description           |
|-----------|------------------------------------------|-----------------------|
| rv.l      | text <b>_[S] label, section</b>   text   | [8t] Section link     |
| rv.l      | text <b>_[D] label, file name</b>   text | [9t] Doc link         |
| rv.l      | text <b>_[U] label, url</b>   text       | [10t] URL link        |
| rv.V, l   | text <b>_[V] var_name</b>   text         | [11t] Variable value  |
| rv.V, l   | <b>text _[E]</b>                         | [12t] Equation number |
| rv.V, l   | <b>title _[T]</b>                        | [13t] Table number    |
| rv.V, l   | <b>caption _[F]</b>                      | [14t] Figure number   |
| rv.V, l   | <b>_[P]</b>                              | [15t] New Page        |
| rv.V, l   | <b>##</b> text                           | nonprinting comment   |

### [3t] Block Tags

#### Format a block of text

| API Scope | Block Tag                             | Description        |
|-----------|---------------------------------------|--------------------|
| rv.R      | <b>_[[SHELL]]</b> process parameters  | [2t] Shell script  |
| rv.l      | <b>_[[INDENT]]</b> spaces (4 default) | [3t] Indent text   |
| rv.l      | <b>_[[ITALIC]]</b> spaces (4 default) | [4t] Indent italic |
| rv.l      | <b>_[[ENDNOTES]]</b> optional label   | [5t] Endnotes      |
| rv.l      | <b>_[[TEXT]]</b> optional language    | [6t] Text          |
| rv.l      | <b>_[[TOPIC]]</b> topic               | [7t] Topic         |
| rv.l      | <b>_[[BOX]]</b> label                 | [8t] Box           |
| rv.V, T   | <b>_[[PYTHON]]</b> namespace          | [9t] Python        |
| rv.D      | <b>_[[METADATA]]</b> label            | [11t] Metadata     |
| rv.D      | <b>_[[LAYOUT]]</b> label              | [12t] Layout block |
| all       | <b>_[[END]]</b>                       | [13t] End block    |

### [4t] Command Summary

#### format files and calculations

| API Scope | Command                                                      | Description          |
|-----------|--------------------------------------------------------------|----------------------|
| rv.R      | <b>SHELL</b>   rel path   os, wait                           | [2t] Shell file      |
| rv.l      | <b>TEXT</b>   rel path   language                            | [3t] Text file       |
| rv.V, l   | <b>TABLE</b>   rel path   title,width,rows,align,head,num    | [4t] Table file      |
| rv.V, l   | <b>IMAGE</b>   rel path   caption, scale, number             | [5t] Image file      |
| rv.V, l   | <b>IMAGE2</b>   rel pth1, rel pth2   cap1,cap2,sca1,sca2,num | [6t] Adjacent images |
| rv.V      | <b>VALTABLE</b>   rel path   title, rows, number             | [7t] Valtable file   |
| rv.V      | a ==: 1*IN   unit1, unit2, decimal   label                   | [8t] Define value    |
| rv.V      | c <=: expression   unit1, unit2, decimal   label, number     | [9t] Assign value    |
| rv.V      | c :=: expression   unit1, unit2, decimal   label, number     | [9t] Function value  |
| rv.V      | a < c   decimal   text1, text2, color1, color2               | [10t] Compare values |
| rv.V, T   | <b>PYTHON</b>   rel path   namespace                         | [11t] Python file    |
| rv.T      | <b>MARKUP</b>   rel path   type                              | [12t] Markup file    |
| rv.D      | <b>ATTACHPDF</b>   rel path   place, title                   | [13t] Attach PDF     |
| rv.D      | <b>PUBLISH</b>   doc title   type                            | [14t] Publish doc    |

### default paths for single docs

See here for the folder structure. If files are in the default path only the file name needs to be provided.

| Command | Default Path |
|---------|--------------|
| SHELL   | folder root  |
| TEXT    | folder root  |

| Command   | Default Path |
|-----------|--------------|
| TABLE     | folder root  |
| IMAGE     | folder root  |
| IMAGE2    | folder root  |
| VALTABLE  | folder root  |
| PYTHON    | folder root  |
| MARKUP    | folder root  |
| ATTACHPDF | folder root  |
| PUBLISH   | folder root  |

### Default command paths for report

See here for the folder structure. If files are in the default path only the file name needs to be provided.

| Command   | Default Path         |
|-----------|----------------------|
| SHELL     | /Files/Scripts/      |
| TEXT      | /Files/Data/         |
| TABLE     | /Files/Data/         |
| IMAGE     | /Files/Image/        |
| IMAGE2    | /Files/Image/        |
| VALTABLE  | /Files/Data/ [1]     |
| PYTHON    | /Files/Scripts/      |
| MARKUP    | /Files/Scripts/      |
| ATTACHPDF | /Files/Attach/       |
| PUBLISH   | depends on file type |

[1] use /stored/data/filename to read values defined in the rivt file

**[5t]** Folders

|                          |                            |
|--------------------------|----------------------------|
| [rivt-]Report-Label/     | Report Folder Name         |
| ├── [.vscode]/           | optional VSCode settings   |
| ├── [rv101-]filename1.py | rivt file                  |
| ├── [rv102-]filename2.py | rivt file                  |
| ├── [rv201-]filename3.py | rivt file                  |
| ├── [rv202-]filename4.py | rivt file                  |
| ├── [_public]/           | public rivt files          |
| ├── rv_101-filename1.py  |                            |
| ├── rv_102-filename1.py  |                            |
| ├── rv_201-filename3.py  |                            |
| ├── rv_202-filename4.py  |                            |
| ├── [_publish]/          | published docs and reports |
| ├── [docs]/              | html output                |
| ├── _images/             |                            |
| ├── _sources/            |                            |
| ├── _static/             |                            |
| ├── process folders/     |                            |
| ├── site folders/        |                            |
| ├── rv101-filename1.html |                            |
| ├── rv102-filename2.html |                            |
| ├── rv201-filename3.html |                            |
| ├── rv202-filename4.html |                            |
| ├── [pdfdocs]/           | pdf doc output             |
| ├── process folders/     |                            |
| ├── rv-101-filename1.pdf |                            |
| ├── rv-102-filename1.pdf |                            |
| ├── rv-201-filename3.pdf |                            |
| ├── rv-202-filename4.pdf |                            |
| ├── [txtdocs]/           | text output                |
| ├── rv-101-filename1.txt |                            |
| ├── rv-102-filename1.txt |                            |
| ├── rv-201-filename3.txt |                            |
| ├── rv-202-filename4.txt |                            |
| ├── [_stored]/           | stored files from rivt     |
| ├── [Logs]/              | log files                  |
| ├── rv101-log.txt        |                            |
| ├── rv102-log.txt        |                            |
| ├── [Sect]/              | stored sections            |
| ├── rv202-5d.txt         |                            |
| ├── rv103-4t.txt         |                            |
| ├── rv301-2r.txt         |                            |
| ├── [Temp]/              | temp files                 |
| ├── rv101-label3.tex     |                            |
| ├── [Vals]/              | stored value files         |
| ├── v101-2.csv           |                            |
| ├── v102-3.csv           |                            |
| ├── [Rst_docs]/          | restructured text files    |
| ├── _downloads/          |                            |
| ├── _static/             |                            |
| ├── _locale/             |                            |

## [6t] Project requirements

|                          |                          |
|--------------------------|--------------------------|
| ├── _templates/          |                          |
| ├── rv101-filename1.rst  |                          |
| ├── rv102-filename2.rst  |                          |
| ├── rv201-filename3.rst  |                          |
| ├── rv202-filename4.rst  |                          |
| ├── gen-html.cmd         | html generating script   |
| ├── gen-pdf.cmd          | pdf generating script    |
| ├── gen-pdftex.cmd       | LaTeX generating script  |
| ├── rivt-report.py       | report generating script |
| ├── new-units.py         | define new units         |
| ├── coverpage.rst        | cover page template      |
| ├── logoname.png         | cover page logo          |
| ├── conf.py              | style paths and settings |
| ├── [Src]                | source files from author |
| ├── data/                | author created subfolder |
| │   ├── data1.csv        |                          |
| │   ├── opensees1.txt    |                          |
| │   └── conc-vals.csv    |                          |
| ├── image/               | author created subfolder |
| │   ├── fig1.png         |                          |
| │   └── fig2.jpg         |                          |
| ├── [Run]/               | OS commands              |
| │   ├── run1_win.cmd     |                          |
| │   └── run1_linux.sh    |                          |
| ├── [Tools]/             | scripts and functions    |
| │   ├── plot.py          |                          |
| │   └── loads.py         |                          |
| ├── [Vals]/              | value files              |
| │   ├── steel-vals.csv   |                          |
| │   └── plastic-vals.csv |                          |
| └── README.txt           | searchable text report   |

## [6t] Project requirements

The minimum Python version is 3.14. A *rivt* project installation includes Python packages for:

- document formatting
- numerical analysis
- symbolic processing
- visualization

| Dependency         | description         |
|--------------------|---------------------|
| "pyzo>=4.20.0"     | lightweight IDE     |
| "pyside6>=6.10.1"  | QT bindings         |
| "fastcore>=1.8.16" | code simplification |
| "tabulate>=0.9.0"  | format tables       |
| "pillow>=11.2.1"   | image processing    |



| Dependency                              | description                  |
|-----------------------------------------|------------------------------|
| "matplotlib">=3.10.1"                   | data visualization           |
| "sympy">=1.13.3"                        | symbolic analysis            |
| "numpy">=2.2.5"                         | numerical analysis           |
| "scipy">=1.16.3"                        | numerical analysis           |
| "pandas">=2.2.3"                        | data analysis                |
| "docutils">=0.21.2"                     | reStructuredText processing  |
| "ipython">=8.16.2"                      | interactive Python shell     |
| "ipykernel">=6.28.1"                    | Jupyter kernel for Python    |
| "reportlab">=4.4.0"                     | PDF generation without LaTeX |
| "rst2pdf">=0.103.1"                     | PDF generation without LaTeX |
| "pypdf">=1.0.3"                         | PDF manipulation             |
| "Sphinx">=8.2.3"                        | HTML generation              |
| "pydata-sphinx-theme">=0.16.1"          | HTML generation              |
| "sphinx-copybutton">=0.5.2"             | HTML generation              |
| "sphinx_design">=0.6.1"                 | HTML generation              |
| "sphinx-favicon">=1.0.1"                | HTML generation              |
| "sphinx-togglebutton">=0.3.2"           | HTML generation              |
| "sphinxcontrib-applehelp">=2.0.0"       | HTML generation              |
| "sphinxcontrib-devhelp">=2.0.0"         | HTML generation              |
| "sphinxcontrib-email">=0.3.6"           | HTML generation              |
| "sphinxcontrib-htmlhelp">=2.1.0"        | HTML generation              |
| "sphinxcontrib-jquery">=4.1"            | HTML generation              |
| "sphinxcontrib-jsmath">=1.0.1"          | HTML generation              |
| "sphinxcontrib-qthelp">=2.0.0"          | HTML generation              |
| "sphinxcontrib-serializinghtml">=2.0.0" | HTML generation              |

## [7t] VSCode settings

Workspace extension and other settings are stored in the `.vscode` folder and can be included and moved as part of a *rivt report*. Settings that affect the *rivt* environment include:

- rivt.code-snippets
- settings.json (extension settings)
- tasks.json
- launch.json

Global settings can be exported and imported using VSCode Profiles. VSCode *rivt* extensions and settings may be installed from this link.

| Extensions                                 | description                |
|--------------------------------------------|----------------------------|
| <b>BUTTONS</b>                             | —                          |
| tombonnike.vscode-status-bar-format-toggle | format button              |
| gsppvo.vscode-commandbar                   | command buttons            |
| AdamAnand.adamstool                        | command buttons            |
| nanlei.save-all                            | save all button            |
| Ho-Wan.setting-toggle                      | toggle settings            |
| yasukotelin.toggle-panel                   | toggle panel               |
| fabiospampinato.vscode-commands            | user command buttons       |
| jerrygoyal.shortcut-menu-bar               | menu bar                   |
| <b>EDITING</b>                             | —                          |
| henryclayton.context-menu-toggle-comments  | toggle comments            |
| TroelsDamgaard.reflow-paragraph            | wrap paragraph             |
| streetsidesoftware.code-spell-checker      | spell check                |
| jmviz.quote-list                           | quote elements in a list   |
| njpwerner.autodocstring                    | insert doc string          |
| oijaz.unicode-latex                        | unicode symbols from latex |
| jsynowiec.vscode-insertdatestring          | insert date string         |
| janisdd.vscode-edit-csv                    | csv editor                 |
| <b>VIEWS</b>                               | —                          |
| GrapeCity.gc-excelviewer                   | excel viewer               |
| SimonSiefke.svg-preview                    | svg viewer                 |
| tomoki1207.pdf                             | pdf viewer                 |
| RandomFractalsInc.vscode-data-preview      | data viewing tools         |

| Extensions                              | description                  |
|-----------------------------------------|------------------------------|
| Fr43nk.seito-openfile                   | open file from path          |
| vikyd.vscod-fold-level                  | line folding tool            |
| file-icons.file-icons                   | icon library                 |
| tintinweb.vscod-inline-bookmarks        | inline bookmarks             |
| <b>MANAGEMENT</b>                       | —                            |
| alefragnani.project-manager             | folder, project management   |
| Anjali.clipboard-history                | clipboard history            |
| dionmunk.vscod-notes                    | notepad                      |
| hbenl.vscod-test-explorer               | test explorer                |
| mightycoco.fsdeploy                     | save file to second location |
| lyzerk.linecounter                      | count lines in files         |
| sandcastle.vscod-open                   | open files in default app    |
| zjffun.snippetsmanager                  | snippet manager              |
| spmeesseman.vscod-taskexplorer          | task explorer                |
| <b>GITHUB</b>                           | —                            |
| GitHub.codespaces                       | run files in codespaces      |
| GitHub.remotehub                        | run remote files             |
| ettoreciprian.vscod-websearch           | search github within VSCode  |
| donjayamanne.githistory                 | git history                  |
| MichaelCurrin.auto-commit-msg           | git auto commit message      |
| github.vscod-github-actions             | github actions               |
| GitHub.vscod-pull-request-github        | github pull request          |
| k9982874.github-gist-explorer           | gist explorer                |
| vsls-contrib.gistfs                     | gist tools                   |
| <b>PYTHON</b>                           | —                            |
| ms-python.autopep8                      | python pep8 formatting       |
| ms-python.isort                         | python sort imports          |
| donjayamanne.python-environment-manager | python library list          |

| Extensions                          | description             |
|-------------------------------------|-------------------------|
| ms-python.python                    | python tools            |
| ms-python.vscode-pylance            | python language server  |
| ms-toolsai.jupyter                  | jupyter tools           |
| ms-toolsai.jupyter-keymap           | jupyter tools           |
| ms-toolsai.jupyter-renderers        | jupyter tools           |
| ms-toolsai.vscode-jupyter-cell-tags | jupyter tools           |
| ms-toolsai.vscode-jupyter-slideshow | jupyter tools           |
| <b>LANGUAGES</b>                    | —                       |
| qwtel.sqlite-viewer                 | sqlite tools            |
| RDebugger.r-debugger                | R tools                 |
| REditorSupport.r                    | R tools                 |
| ms-vscode-remote.remote-wsl         | windows linux tools     |
| James-Yu.latex-workshop             | latex tools             |
| lextudio.restructuredtext           | restructured text tools |
| trond-snekvik.simple-rst            | restructured syntax     |
| yzane.markdown-pdf                  | markdown to pdf         |
| yzhang.markdown-all-in-one          | markdown tools          |

| Snippets/Keys | description                                        |
|---------------|----------------------------------------------------|
| run           | API Run function                                   |
| ins           | API Insert function                                |
| val           | API Values function                                |
| too           | API Tools function                                 |
| wri           | API Write function                                 |
| alt+q         | rewrap paragraph with hard line feeds (80 default) |
| alt+p         | open file under cursor                             |
| alt+.         | select correct spelling under cursor               |
| alt+8         | insert date                                        |

| Snippets/Keys | description                               |
|---------------|-------------------------------------------|
| alt+9         | insert time                               |
| ctl+1         | focus on first editor                     |
| ctl+2         | focus on next editor                      |
| ctl+3         | focus on previous editor                  |
| ctl+8         | focus on explorer pane                    |
| ctl+9         | focus on github pane                      |
| ctl+alt+x     | reload window                             |
| ctl+alt+[     | reload window                             |
| ctl+alt+]     | unfold all code                           |
| ctl+alt+u     | unfold all code                           |
| ctl+alt+f     | fold code level 2 (rivt sections visible) |
| ctl+alt+a     | fold code - all levels                    |
| ctl+alt+t     | toggle local fold                         |
| ctl+alt+e     | toggle explorer sort order                |
| ctl+alt+s     | toggle spell check                        |
| ctl+alt+g     | next editor group                         |
| ctl+shift+u   | open URL under cursor in browser          |
| ctl+shift+s   | open GitHub README search for rivt        |
| ctl+shift+a   | commit all                                |
| ctl+shift+z   | commit the current editor                 |
| ctl+shift+x   | post to remote                            |

| Keystroke | Description                                        |
|-----------|----------------------------------------------------|
| alt+q     | rewrap paragraph with hard line feeds (80 default) |
| alt+p     | open file under cursor                             |
| alt+.     | select correct spelling under cursor               |
| alt+8     | insert date                                        |
| alt+9     | insert time                                        |

| Keystroke   | Description                               |
|-------------|-------------------------------------------|
| ctl+1       | focus on first editor                     |
| ctl+2       | focus on next editor                      |
| ctl+3       | focus on previous editor                  |
| ctl+8       | focus on explorer pane                    |
| ctl+9       | focus on github pane                      |
| ctl+alt+x   | reload window                             |
| ctl+alt+u   | unfold all code                           |
| ctl+alt+f   | fold code level 2 (rivt sections visible) |
| ctl+alt+a   | fold code - all levels                    |
| ctl+alt+t   | toggle local fold                         |
| ctl+alt+e   | toggle explorer sort order                |
| ctl+alt+s   | toggle spell check                        |
| ctl+alt+g   | next editor group                         |
| ctl+shift+u | open URL under cursor in browser          |
| ctl+shift+s | open GitHub rivt README search            |
| ctl+shift+a | commit all                                |
| ctl+shift+z | commit current editor                     |
| ctl+shift+x | post to remote                            |

## A.2 Terms

### [1t] rivt Terms

#### block tag : block tags

A block tag formats a block of text that begins with `_[[TAG]]` and terminates with `_[[Q]]`.

#### command : commands

*riv* commands read and write external files and assign values to variables. They start in the initial columns with a vertical bar ( | ) followed by the file path, name and parameters. The path is relative to the rivt report root folder. If the metadata variable *rv\_localB* is set to true resources will be read from and written to the local *riv* file folder.

```
| COMMAND | relative path | parameters
```

#### content : content text : content substring

The text in a *rivt string* that follows the *header substring*. The text is utf-8 (Unicode Transformation Format - 8-bit), a character encoding standard that is a superset of ASCII, handles text from any language, and is constant across all platforms.

### **division**

A group of related rivt files organized in a report and labeled in the rivt file name. Optionally organized in a division folder.

### **doc : docs**

A formatted document file output using the rivtlib package. A doc is a text, PDF or HTML file with the default name of the *rivt file* and file type suffix.

### **doc number**

Prefix of a rivt and doc file name and used to organize a report. It has the form *rvDss-filename.py* where D is a capital alphanumeric division number and ss is the subdivision number.

### **header : headers : header text : header substring**

The first line of a *rivt string* that includes a section title, followed by comma separated parameters.

### **line tag : line tags**

A line tag has the form `_[TAG]` and formats a line of text.

### **report : reports : rivt report : rivt reports**

A group of compiled *docs* organized by rivt file number.

### **report folder : report folders**

The folder structure for producing a report is described here.

### **report script**

A Python script that assembles *docs* into a *report*.

### **rivt**

An open source Python project that includes rivtlib and approximately two dozen Python packages. When *rivt* is installed docs and reports may be edited and published using a text editor.

### **rivt doc : rivt docs**

A formatted document file output from a *rivt file* using the rivtlib package. A doc is a text, PDF or HTML file with the default name of the *rivt file* and file type suffix.

### **rivt file : rivt files**

A Python file that imports *rivtlib* and includes *rivt markup*.

### **rivt file number**

Prefix of rivt file name used to organize a report (rvDss-). It has the form *rvDss-filename.py* where D is a capital alphanumeric division number and ss is the subdivision number.

### **rivt folder**

A folder containing a rivt file and all of its resources in the same folder. Generally used

### **rivt framework : framework**

The recommended *rivt-framework* includes the following separately installed programs:

1. VSCode <<https://code.visualstudio.com/>> and VSCode extensions <<https://marketplace.visualstudio.com/vscode>>. rivt extensions and snippets are here.
2. Git <<https://git-scm.com>> and set up a GitHub <<https://github.com>>
3. LaTeX <<https://www.tug.org/texlive/>>
4. QCAD <<https://qcad.io/en/>>

### **rivt log file**

*rivt* file execution log written to the *log* folder as *rvDss-log.txt*.

**rivt markup**

A light weight markup that wraps restructuredText and outputs reports in text, PDF and HTML. Markup is included in string arguments to rivt API functions.

**rivt report folder**

A folder containing multiple rivt files to be organized in a report and resources organized in subfolders

**rivt string**

Triple quoted string argument to an API function.

**rivtlib**

Open source [Python package](#) that generates docs and reports from a rivt file

**rvspace**

rivt alias name for the global rivt file namespace. Other user namespaces may be specified

**section parameter : section parameters**

Comma separated parameters in a *header* that specify the section processing.

**single doc : single docs**

A single doc is a rivt file that publishes a doc that is not part of a report and is used for quick docs that only require limited formatting.

It is specified in a comment variable directly after the *rivtlib* import statement.

```
# rv singledoc = True
```

The default setting is False. The rivt file folder is used for source and output files. A folder structure is not used.

**[2t] Python Terms**

**docutils**

A Python package that processes [restructured text](#) files into HTML, LaTeX, and other formats.

**namespace**

Provides [scope](#) for functions and variables.

**Python : python**

A [programming language](#) that lets you work quickly and integrate systems more effectively

**restructuredtext : restructuredtext markup**

A lightweight markup language designed to be processed by document software including [docutils](#), Sphinx and rivt.

**[3t] GitHub and VSCode Terms**

**fork : forked**

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. Most commonly, forks are used to either propose changes to someone else's project or to use someone else's project as a starting point for your own idea.

**Git : git**

A [version control system](#) that lets you manage and keep track of your source code history.



**IDE : ide**

Integrated Development Environment. A software application that provides comprehensive facilities to computer programmers for software development. An IDE typically includes a code editor, build automation tools, and a debugger. *rivt extensions* have been developed for VSCode. Other IDEs include PyCharm, Spyder, and JupyterLab.

**profile**

Allows VSCode users to customize their environment for different workflows, projects, or tasks. This feature provides a way to manage distinct configurations of settings, extensions, keyboard shortcuts, snippets, and tasks.

**repo : repository**

a storage location for software packages

**rivt Codespace : rivt codespace**

A Codespace set up with extensions, snippets and keyboard shortcuts for editing and publishing *rivt docs*.



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